Chapter 10

Monte Carlo Simulation of Neutral Particle Transport

Particle transport calculations
do not always meet expectations.
The trouble, of course,
is using brute force
rather than variance reductions.

Numerical solutions of the transport equation, called *deterministic* calculations, usually require a discretization of the time, energy, angular, and spatial variables. Each such approximation introduces a *systematic* error. Moreover, the geometry for deterministic calculations must be rather simple, for example, planar, cylindrical, or spherical surfaces, and a full three-dimensional calculation requires tremendous computer memories to hold all the discretized values of the energy and angular flux density.

Monte Carlo calculations, by contrast, are capable of treating very complex three-dimensional geometries. The continuous treatment of time, energy, direction, and space avoids all systematic discretization errors. However, the stochastic simulation of the transport process introduces *statistical* errors in the results, errors which are not present in deterministic calculations. Historically, Monte Carlo analyses could be performed only on large mainframe computers because of the intensive computational requirements. But today's inexpensive personal computers are capable of running very complex Monte Carlo simulations.

Monte Carlo calculations are not always better than deterministic approaches. Monte Carlo excels at estimating some quantity averaged over, at most, a few regions of phase space $\boldsymbol{P} = (\mathbf{r}, E, \boldsymbol{\Omega}, t)$. But, if, for example, detailed three-dimensional spatial profiles of the flux in a reactor core are needed, deterministic methods often are superior. The reason for this is that, to obtain with Monte Carlo the value of the flux density at a large number of points, the spatial domain would have to be broken into a vast number of small contiguous volumes ΔV and the flux density estimated in each of these volumes. The number of histories that contribute to the flux estimates in the volumes decrease rapidly as ΔV is made smaller to increase the spatial resolution, thereby causing the statistical uncertainty in the estimated flux densities to increase, sometimes

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to unacceptable values. However, Monte Carlo techniques are, or can be made, efficient when the number of results or tallies are small.

10.1 Basic Approach for Monte Carlo Transport Simulations

A Monte Carlo radiation transport calculation simulates a finite number N of particle histories by sampling from the appropriate probability density functions (PDFs) governing the various events that may happen to a particle from its birth by a source to its eventual demise by being absorbed or escaping through the problem boundary. A history begins by randomly selecting a particle's initial position, energy, and direction from the PDFs that describe the sources for the problem. As the particle travels on various legs of its trajectory, sampling is performed to determine random values for distance to the next interaction, type of interaction, scattering angle, energy change, and so on. Between interactions, the neutrons or photons travel in straight lines. On some interactions, secondary particles are created, such as neutrons from fission reactions or X-rays from photon photoelectric interactions. These secondary particles are "banked" and their initial energies, interaction positions, and travel directions are recorded. After the original particle finishes its history, these secondary banked particles are processed sequentially to create a history for each.

After each history is terminated (or after its interactions or surface crossings are determined), its contribution z_i is added to the tally being used to estimate the quantity of interest, for example, average flux in a cell, probability of escape, energy deposited in a cell, current through a surface, or many other quantities related to the radiation field. After N histories, the estimate of $\langle z \rangle$ is

$$\langle z \rangle \simeq \overline{z} = \frac{1}{N} \sum_{i=1}^{N} z_i .$$

Along with such an estimate, various measures of its statistical reliability must be made (see Section 4.2).

Such an analog simulation mimics the stochastic events that befall an actual particle if the problem was converted to the equivalent experiment. In many such calculations, extremely few of the histories contribute anything to the tally of interest. For example, if the probability a neutron incident on a thick shield eventually reaches the other side is 1 in 10^9, then in an analog simulation only one out of 10^9 histories, on average, terminates on the back side of the shield. For such problems, it is vital to use variance reduction techniques (see Chapter 5) and perform *nonanalog* Monte Carlo calculations.

10.2 Geometry

Almost all Monte Carlo codes use a Cartesian coordinate system, even if the problem possesses cylindrical or spherical symmetry. The reason for this is that, as a particle travels along the direction $\mathbf{\Omega} = \mathbf{i}u + \mathbf{j}v + \mathbf{k}w$, the direction cosines $u = \mathbf{i}\bullet\mathbf{\Omega}$, $v = \mathbf{j}\bullet\mathbf{\Omega}$, and $w = \mathbf{k}\bullet\mathbf{\Omega}$ with respect to the coordinate axes do not change. If a particle moves a distance s from $\mathbf{r}_0$ in direction $\mathbf{\Omega}$, its new position $\mathbf{r}$ is

$$\mathbf{r} = \mathbf{r}_0 + s\mathbf{\Omega} = (x_0 + su, y_0 + sv, z_0 + sw). \tag{10.1}$$

This simplicity in describing a particle's position as it moves in a straight line is not possible in other coordinate systems.

The three-dimensional space in which the simulated histories are constructed is usually assumed to be composed of contiguous homogeneous volumes or *cells*, each cell being bounded by one or more surfaces (or portions of surfaces). A cell can be a void or composed of any homogeneous material. All of three-dimensional space must belong to some cell—there can be no "holes" where a particle would become "lost." Although not necessary, the geometry of the problem is often surrounded by a *problem boundary*, usually far from tally regions. The cell containing all space beyond the problem boundary is the "graveyard," whose purpose is to "kill" any particle that enters it, thereby saving time by not tracking particles beyond the boundary and that have negligible chance of contributing to the tally. For example, in calculating the doses to a patient from an X-ray therapy unit, the walls of the treatment room should be modeled, but nearby buildings and airplanes passing overhead have little chance of affecting the dose to the patient. Similarly, objects far from tally regions often can be modeled crudely compared to objects near the important regions. For example, door knobs in the treatment room need not be modeled unless the problem is to determine how much radiation leaks through the key hole. Constructing an effective but efficient geometry model for a particular problem is a matter of skill and experience on the modeler's part.

10.2.1 Combinatorial Geometry

Surfaces are usually defined by first-degree functions (planes) or second-degree functions (cylinders, cones, ellipsoids, etc.) as $f(x, y, z) = 0$. Sometimes fourth-degree functions (toroids) are used, although such surfaces present greater numerical difficulties for evaluating the intersection point with a particle trajectory. Commonly used surfaces are given in Table 10.1. Surface A, defined by $f_A(x, y, z) = 0$, divides all space into two subdomains, namely, those points $\mathbf{r} = (x, y, z)$ for which $f_A(x, y, z) > 0$ (denoted by A^+) and those points for which $f_A(x, y, z) < 0$ (denoted by A^-). For example, consider the spherical surface $S : f_S(x, y, z) = x^2 + y^2 + z^2 - R^2 = 0$. All points inside the sphere (S^-) are to the negative side of the surface, while all points outside (S^+) are to the positive side of the surface.

By using Boolean unions (∪ "and") and intersections (∩ "or") of the two subdomains for each surface, volumes of quite complex shapes can be approximated. As a simple example, consider the construction of a finite cylinder of

Table 10.1 Some first-, second-, and fourth-degree surfaces. The location and orientation of the surfaces are determined by the values of the surface parameters denoted by capital and Greek letters. Extracted from MCNP (2003).

Type	Description	Surface $f(x,y,z)=0$
plane	general	$Ax+By+Cz-D=0$
	normal to x-axis	$x-D=0$
	normal to y-axis	$y-D=0$
	normal to z-axis	$z-D=0$
sphere	centered at origin	$x^2+y^2+z^2-R^2=0$
	general	$(x-A)^2+(y-B)^2+(z-C)^2-R^2=0$
	centered on x-axis	$(x-A)^2+y^2+z^2-R^2=0$
	centered on y-axis	$x^2+(y-B)^2+z^2-R^2=0$
	centered on z-axis	$x^2+y^2+(z-C)^2-R^2=0$
cylinder	parallel to x-axis	$(y-B)^2+(z-C)^2-R^2=0$
	parallel to y-axis	$(x-A)^2+(z-C)^2-R^2=0$
	parallel to z-axis	$(x-A)^2+(y-B)^2-R^2=0$
	on x-axis	$y^2+z^2-R^2=0$
	on y-axis	$x^2+z^2-R^2=0$
	on z-axis	$x^2+y^2-R^2=0$
cone	parallel to x-axis	$(y-B)^2+(z-C)^2-D(x-A)^2=0$
	parallel to y-axis	$(x-A)^2+(z-C)^2-D(y-B)^2=0$
	parallel to z-axis	$(x-A)^2+(y-B)^2-D(z-C)^2=0$
	on x-axis	$y^2+z^2-D(x-A)^2=0$
	on y-axis	$x^2+z^2-D(y-B)^2=0$
	on z-axis	$x^2+y^2-D(z-C)^2=0$
ellipsoid, hyperboloid, paraboloid	axis parallel to x-, y-, or z-axis	$A(x-\alpha)^2+B(y-\beta)^2+C(z-\gamma)^2$ $+2D(x-\alpha)+2E(y-\beta)$ $+2F(z-\gamma)+G=0$
cylinder, cone, ellipsoid, paraboloid, hyperboloid	axis not parallel to x-, y-, or z-axis	$Ax^2+By^2+Cz^2+Dxy+Eyz$ $+Fzx+Gz+Hy+Jz+K=0$
elliptical or circular torus. Axis is parallel to x-, y-, or z-axis	$(x-\alpha)^2/B^2+(\sqrt{(y-\beta)^2+(z-\gamma)^2}-A)^2/C^2-1=0$	
	$(y-\beta)^2/B^2+(\sqrt{(x-\alpha)^2+(z-\gamma)^2}-A)^2/C^2-1=0$	
	$(z-\gamma)^2/B^2+(\sqrt{(x-\alpha)^2+(y-\beta)^2}-A)^2/C^2-1=0$	

radius R and height $2H$, centered at the origin, and with its axis on the x-axis. Such a cylindrical volume is bounded by two parallel planes perpendicular to the x-axis, $f_{P_1}(x, y, z) \equiv x - H = 0$ and $f_{P_2}(x, y, z) \equiv x + H = 0$, and the infinite cylinder $f_C(x, y, z) \equiv y^2 + z^2 - R^2 = 0$. The space inside this cylinder is specified as

$$P_1^- \cap P_2^+ \cap C^-,$$

while all of the space outside this finite cylinder is described as

$$P_1^+ \cup P_2^- \cup C^+.$$

To permit rapid particle tracking through the geometry, each region (or cell) and each surface are uniquely numbered. Then, for each cell, the numbers of the surfaces bounding it are tabulated as are the specifications of the regions on the other side of the bounding surfaces. Although somewhat redundant, this bookkeeping makes tracking particles very efficient because when a particle leaves a region, the adjacent regions are immediately known as are the new surfaces toward which the particle is traveling.

10.3 Sources

Each particle history is begun by sampling from the spatial, energy, and angular distributions of the source to determine the starting position, energy, and direction of each particle history. The source may be specified explicitly (a *fixed source* problem) or calculated by the simulation itself (an *eigenvalue* problem). In the latter case, initial simulations are used to estimate the spatial distribution of sources such as a fission source and the resulting equilibrium distribution in a critical system. However, in this chapter only the fixed source problem is discussed.

Sources in a transport calculation can be distributed over many regions, be localized to a few cells, or be singular sources such as point, line, and plane sources. The energy of particles emitted by the sources can consist of a discrete set of energies (typical of gamma rays produced by radioactive decay) or have a continuous distribution of energies (such as neutrons produced by spontaneous fission). While many sources emit radiation isotropically, some sources can be anisotropic (such as a beam of radiation from a reactor beam port). To start a particle history, one must sample from the specified spatial, energy, and angular distributions. To complicate matters, these three distributions are often interrelated. For example, different source regions may emit particles with different energy and angular distributions. Modeling the source distributions can often be quite a complex task.

10.3.1 Isotropic Sources

Many physical sources emit radiation isotropically. The probability $p(\mathbf{\Omega})d\Omega$ a source particle is emitted in $d\Omega$ about a direction $\mathbf{\Omega}$ is thus

$$p(\mathbf{\Omega})d\Omega = \frac{d\Omega}{4\pi} = \frac{d\psi}{2\pi}\frac{d\omega}{2}, \tag{10.2}$$

where $\omega = \cos\theta$. Thus, the PDF of the azimuthal angle is seen to be $p(\psi) = 1/(2\pi)$, $0 \le \psi < 2\pi$, and the PDF for the cosine of the polar angle θ is $p(\omega) = 1/2$, $-1 \le \omega \le 1$, both uniform distributions. Sampling from these distributions gives

$$\psi_i = 2\pi\rho_j \qquad \text{and} \qquad \omega_i = -1 + 2\rho_{j+1}, \tag{10.3}$$

where ρ_j and ρ_{j+1} are two random numbers uniformly distributed between (0, 1). The directed cosines of the initial direction $\mathbf{\Omega}_i$ of a source particle are then calculated as

$$u_i = \mathbf{i}\bullet\mathbf{\Omega} = \cos\psi_i\sqrt{1-\omega_i^2}, \quad v_i = \mathbf{j}\bullet\mathbf{\Omega} = \sin\psi_i\sqrt{1-\omega_i^2}, \quad w_i = \mathbf{k}\bullet\mathbf{\Omega} = \omega_i. \tag{10.4}$$

10.4 Path Length Estimation

A key component for calculating a particle history is a geometry module that can identify for a particle at $\mathbf{r}_0$ (1) what surface (if any) it will first hit if it travels a distance s in direction $\mathbf{\Omega}$, (2) what is the distance to the intersection point if it hits the surface, and (3) what region or cell is on the other side.

10.4.1 Travel Distance in Each Cell

The distance a particle travels before interacting is a random variable. The path length s of each leg of a particle's history is estimated from its PDF (see Eq. (9.27)) $f(s) = \mu e^{-\mu s}$, where μ is the total interaction coefficient. Because μ generally varies from region to region and because a track segment may span multiple regions, the PDF for travel distance is expressed in terms of mean free path lengths $\lambda \equiv \mu s$. Thus, the distance to the next collision, in mean free path lengths, has a PDF $f(\lambda) = e^{-\lambda}$ and is independent of the medium. The length of each track segment is then obtained by sampling from an exponential distribution (see Example 4.2) as $\lambda = -\ln\rho$, where ρ is from $\mathcal{U}(0, 1)$.

Let lengths $s_1, s_2, \ldots, s_n$ be the segments of the flight path in each region along the particle's direction of travel (see Fig. 10.1). Then, if a particle starts at $\mathbf{r}_0$ and if

$$\sum_{i=1}^{n-1} \mu_i s_i \le \lambda < \sum_{i=1}^{n} \mu_i s_i, \tag{10.5}$$

the next collision occurs in region n at a distance

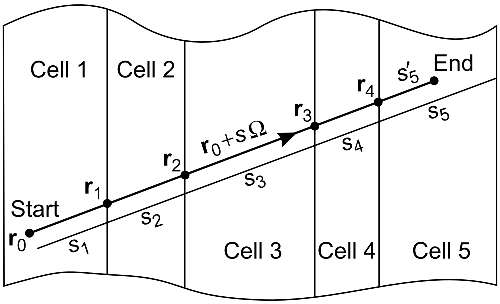


Figure 10.1 Path lengths in each cell along a straight-line segment of a particle trajectory.

$$s'_n = \frac{1}{\mu_n}\left(\lambda - \sum_{i=1}^{n-1} \mu_i s_i\right) \tag{10.6}$$

beyond the entrance point to the nth region. Here μ_i is the interaction coefficient for the ith region.

The distances s_i are readily calculated from the intersection of the straight-line trajectory of Eq. (10.1) with the surfaces $f_i(x, y, z)$ through which it passes. The straight line, along which the particle moves, can alternatively be written as

$$\mathbf{r} = \mathbf{r}_{i-1} + \mathbf{\Omega} s, \qquad i = 1, 2, \ldots, n, \tag{10.7}$$

where $\mathbf{r}_{i-1}$ is the position where the particle begins its flight in region i. Then the distance to the point of exit s_i from this region is found by (1) first solving

$$f_m^i(x_{i-1} + us_{i_m}, y_{i-1} + vs_{i_m}, z_{i-1} + ws_{i_m}) = 0, \tag{10.8}$$

where f_m^i, $m = 1, 2, \ldots$, are all the surfaces bounding region i, and (2) then by picking s_i as the smallest positive real solution s_{i_m}, $m = 1, 2, \ldots$. If $f_m^i(x, y, z)$ are quadratic surfaces,[1] then Eq. (10.8) is a quadratic equation for s_{i_m}, namely,

$$as_{i_m}^2 + 2bs_{i_m} + c = 0, \tag{10.9}$$

where a, b, and c depend on the parameters of f_m^i and on $(x_{i-1}, y_{i-1}, z_{i-1})$. Of course, in performing this calculation for all the surfaces bounding region i, most of the double roots s_{i_m} will be imaginary (meaning no intersection) or negative (an intersection in the backward direction). The smallest positive real root gives the path distance s_i in region i.

[1] The most general form is $f(x, y, z) = Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz + J$. The 10 parameters A–J define 10 possible surfaces: ellipsoids, cones, cylinders, hyperboloids of one sheet, hyperboloids of two sheets, elliptic paraboloids, hyperbolic cylinders, hyperbolic paraboloids, parabolic cylinders, and planes (Olmsted, 1947).

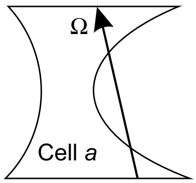


Figure 10.2 A concave cell whose concavities are single surfaces.

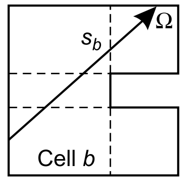


Figure 10.3 A concave cell whose concavities are from multiple surfaces.

10.4.2 Convex Versus Concave Cells

So far, it has been tacitly assumed that the various regions are all convex, i.e., a straight line between any two points lies wholly within the region. Actually, concavities in a cell due to single surfaces (see Fig. 10.2) present no difficulties with the previous procedure for calculating path lengths in a cell. However, if the concavities are produced by multiple surfaces (the dashed lines in Fig. 10.3), additional tests are required to eliminate irrelevant surface intersections by determining if the same cell is on the other side of these spurious intersections.

10.4.3 Effect of Computer Precision

Although the previous description of how a particle is traced along its trajectory of straight-line segments, punctuated by point interactions, is conceptually simple, its implementation in computer code is not so easy because of the finite precision with which numbers can be represented. The position $\mathbf{r}_i$ of the intersection of a trajectory with surface f_i, bounding the cells i and $i+1$, may be rounded by the finite precision of the computer to (1) still remain in region i, (2) be placed in cell $i+1$, or (3) actually be on the surface (see Fig. 10.4). In all cases, the computer "thinks" the particle is entering cell $i+1$.

The calculation of the distance s_{i+1} of the path length in the next cell $i+1$ is done by finding the smallest positive distance to the surfaces bounding cell $i+1$ (here surfaces f_i and f_{i+1}). All three cases give essentially the correct distance to surface f_{i+1}; however, the distance to f_i is a small positive value for case 1 (undershoot), a small negative value for case 2 (overshoot), and zero for case 3 (exact).[2] Only in cases 2 and 3 is the correct distance s_{i+1} to the exit point in cell $i+1$ obtained.

It is tempting to add a small positive distance to s_i to ensure there are never any undershoots. But such a fix prevents the code from being "scale-invariant," i.e., the size of the correction depends on the size of the geometry. Also use of higher-precision arithmetic does not fix this problem, but makes the under/overshoots only smaller. Fortunately there is a simple remedy to avoid these problems.

[2] Round-off also affects these small or zero distances to f_i, and any one of them could be rounded into a small positive distance and, thus, gives the incorrect distance to the exit point in cell $i+1$.

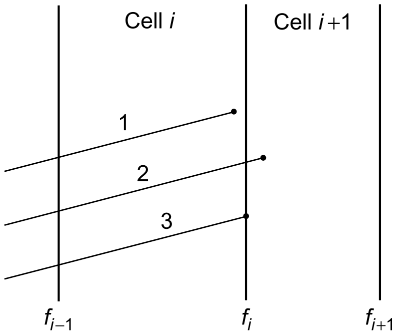


Figure 10.4 Three possibilities when a particle is transported to surface f_i.

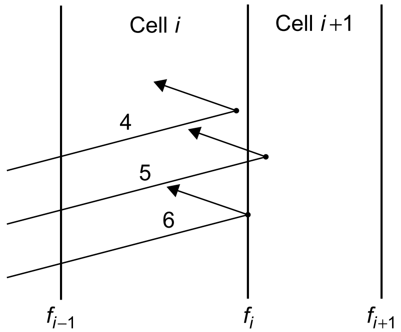


Figure 10.5 Three possibilities when a particle is transported to and reflected from surface f_i.

Rule 1. *Do not calculate distances to boundaries from which the particle is receding based on its known direction* $\mathbf{\Omega}$ *and cell the particle thinks it is in. This resolves all undershoot problems and small or zero distances to surfaces.*

A further complicating factor occurs if the particle changes direction on reaching the boundary. This change in direction occurs if the boundary is a "reflecting" boundary (used, for example, to take advantage of problem symmetries) or if (very rarely) the next interaction is on a boundary and is a scattering event. The ambiguities on where near the boundary the particle is and a possible change in its direction can cause serious errors in the calculation of the next leg of the trajectory. In Fig. 10.5, the particle has been transported to surface f_i, its direction reversed, but the computer still thinks it is entering region $i + 1$. In the next step, only the distance to surface f_i is made because the particle is receding from surface $i + 1$. Cases 4, 5, and 6 return a small negative, a small positive, and a zero value, respectively. Such problems are resolved by using another tracking rule.

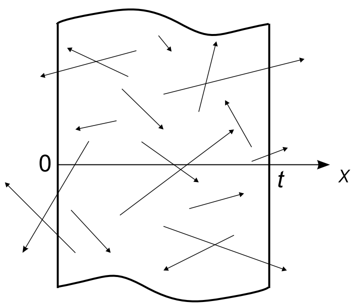
Rule 2. *For a flight distance that is negative or zero, use a zero step size and make a region change. This resolves all undershoot or exact intersections when the particle is backscattered at the surface.*

Finally, it should be mentioned that, with finite-precision arithmetic, it is possible in complex geometries for a particle to become "lost." For example, if several cells have a common vertex, a particle reaching the vertex (within machine precision) may not be able to resolve which region it is entering on the other side of the vertex. Such problems, however, are extremely rare (the probability any trajectory exactly passes through any specified point is zero). For problems using millions of histories, one or two "lost" particles have negligible effect on the results.

10.5 Purely Absorbing Media

A degenerate form of transport theory is to calculate the probability that a neutron born in a region escapes without an interaction. In essence, each collision is treated as an absorption, and the problem is to estimate the probability of escape from the region without undergoing any collision. In other words, what is the fraction of the particles born in the region that have a path length to the first collision site that lies outside the region? For this special case, exact analytic solutions for simple geometries are known (Case et al., 1953). More important, such escape-probability calculations are readily evaluated with the Monte Carlo method using only track length simulation. Such a calculation is given in Example 10.1 and compared to the analytical result.

Example 10.1 Escape Probability From an Infinite Slab



A simple application of sampling path lengths of neutral particles is the determination of the escape probability P_{esc} of particles emitted uniformly in some purely absorbing body, for which $\mu = \mu_a$. For simple bodies, such as infinite slabs, spheres, and infinite cylinders, analytical solutions of the transport equation Eq. (9.50) are known (Case et al., 1953). As an example, consider the case in which particles are emitted uniformly and isotropically throughout an infinite absorbing slab of thickness t as shown in the figure. Because the slab is infinite in the x- and y-directions, only the x-position where the particle is born affects its chance of escape. Further, for isotropic emission, the cosine of the emission angle θ with respect to the positive polar x-axis is uniformly distributed in $(-1, 1)$. The azimuthal angle of emission does not affect the escape probability, so only the polar angle θ need be simulated.

The following Monte Carlo game is played N times to determine the number of simulated particles that escape, N_{esc}.

1. Select an emission point uniformly in $(0, t)$ as $x_i = t\rho_i$.
2. Select an emission direction $\omega \equiv \cos\theta$ as $\omega_i = 2\rho_i - 1$.
3. Calculate the distance d to the surface. If $\omega_i > 0$, then $d_i = (t - x_i)/\omega_i$; otherwise $d_i = x_i/\omega_i$.
4. Generate a random path length s for the particle's first collision (absorption) in the material from the distribution $f(s) = \mu e^{-\mu s}$ as (see Example 4.2) $s_i = [-\ln \rho_i]/\mu$.
5. If $s_i < d_i$, the particle is absorbed and $z_i = 0$; otherwise the particle escapes and $z_i = 1$.

After performing this simulation N times the escape probability is estimated from $\overline{z} = (1/N)\sum_{i=1}^{N} z_i$ (which also equals $\overline{z^2} = (1/N)\sum_{i=1}^{N} z_i^2$) as

$$P_{esc} \simeq \overline{z} \pm \frac{1}{\sqrt{N}}\left[\overline{z^2} - \overline{z}^2\right].$$

The analytic solution for this problem is (Case et al., 1953)

$$P_{esc} = \mu t\left[\frac{1}{2} - E_3(\mu t)\right],$$

where $E_3(x) \equiv x^2\int_x^\infty u^{-3}e^{-u}\,du$ is the exponential integral function of order 3. Results using $N = 1000$ particles are compared to the analytical result for several slab widths in Fig. 10.6.

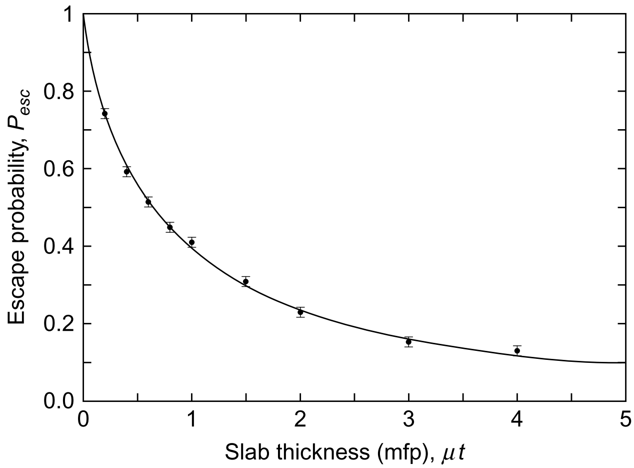


Figure 10.6 The escape probability P_{esc} for a purely absorbing slab with a uniform isotropic source of particles. The line is the analytical result, and the data are Monte Carlo simulation results using 1000 histories for each point. Error bars are the estimated standard deviations.

10.6 Type of Collision

At the end of a straight-line path segment, the particle undergoes an interaction with the material in the current cell.[3] The probability that the particle with energy E interacts with species j (an isotope for neutrons or an element for photons) in a reaction of type i is

$$p_i^j = \frac{\mu_i^j(E)}{\sum_i \sum_j \mu_i^j(E)}, \tag{10.10}$$

[3] Of course, if the collision site is outside the problem boundary, the history is terminated, tallies are updated, and a new source particle history is started.

where the summations are over all species in the cell and over all possible reaction types.

If the reaction is one in which no particle of the type being tracked emerges from the interaction, the particle is absorbed and the history ends. If more than one particle of the type being considered results from the interaction (e.g., fission for neutrons or fluorescence and X-rays for photons), the energy and direction of all but one are "banked" for later processing and one particle continues the presented history.

10.6.1 Scattering Interactions

Scattering is a frequent type of interaction in which the incident particle is deflected into a new direction $\mathbf{\Omega}'$ from its old direction $\mathbf{\Omega}$ through a scattering angle $\theta_s = \cos^{-1} \mathbf{\Omega}\bullet\mathbf{\Omega}$. In a scattering interaction with species j, the constraints of conservation of energy and momentum force the doubly differential scattering interaction coefficient (see Section 9.2.4) $\mu_s^j(E \to E', \mathbf{\Omega}\bullet\mathbf{\Omega}')$ to be proportional to $\delta(\mathbf{\Omega}\bullet\mathbf{\Omega}' - S(E, E'))$, i.e., if any two of the variables E, E', and $\cos\theta_s = \mathbf{\Omega}\bullet\mathbf{\Omega}'$ are specified, the third is forced to have a value that makes the argument of the delta function zero. Thus, the scattering angle $\cos\theta_s$ is sampled (usually with the rejection technique) from the PDF formed from the empirical[4] singly differential scattering interaction coefficient

$$f(\mathbf{\Omega}\bullet\mathbf{\Omega}') = \frac{\mu_s^j(E, \mathbf{\Omega}\bullet\mathbf{\Omega}')}{\int_{-1}^{1} \mu_s^j(E, \mathbf{\Omega}\bullet\mathbf{\Omega}')\, d(\mathbf{\Omega}\bullet\mathbf{\Omega}')}. \tag{10.11}$$

Once $\cos\theta_s$ is determined, the new energy of the particle is obtained by solving

$$\mathbf{\Omega}\bullet\mathbf{\Omega}' = \cos\theta_s = S(E, E') \tag{10.12}$$

for E'.

Direction Cosines of the Scattered Particle

Once the scattering angle and energy of the scattered particle are determined, the direction cosines (u', v', w') of $\mathbf{\Omega}'$ need to be calculated. The geometry is illustrated in Fig. 10.7. The change in azimuth $\Delta\psi_s = \psi' - \psi$ is uniformly distributed in $(0, 2\pi)$ radians. Let θ and ψ be the polar and azimuthal angles before scatter, with direction cosines $u = \sin\theta\cos\psi$, $v = \sin\theta\sin\psi$, and $w = \cos\theta$.

To calculate the direction cosines (u', v', w') of $\mathbf{\Omega}'$ it is first necessary to introduce rotation matrices $\mathbf{R}$. Consider first a rotation about the z-axis through an angle ψ, as shown in Fig. 10.8. The coordinates of the point P in original

[4] Only for the case of Compton scattering of photons is there an analytical expression for the differential scattering interaction coefficient.

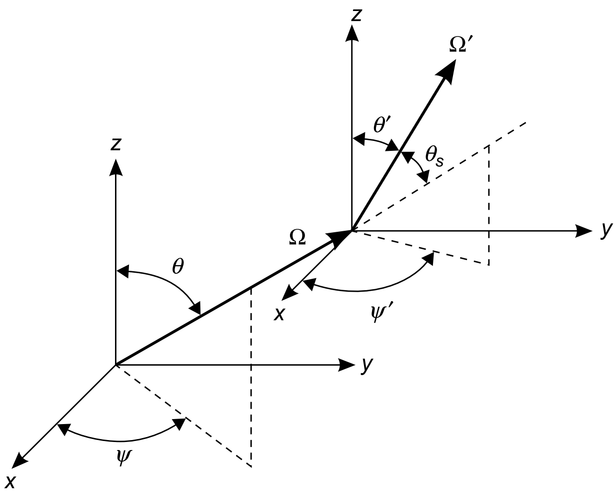


Figure 10.7 The relation between the coordinates of the particle in direction $\mathbf{\Omega}$ before a scatter and the direction $\mathbf{\Omega}'$ after the scatter. The angle between $\mathbf{\Omega}$ and $\mathbf{\Omega}'$ is the scattering angle θ_s, namely, $\cos\theta_s = \mathbf{\Omega} \bullet \mathbf{\Omega}'$.

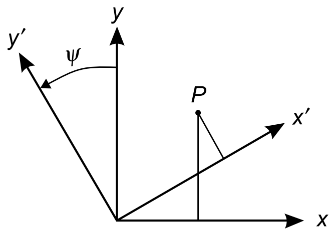


Figure 10.8 A rotation in the positive sense about the z-axis.

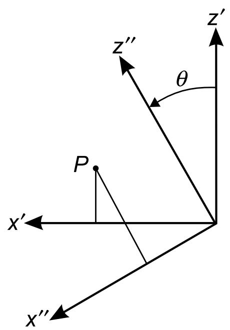


Figure 10.9 A rotation about the y-axis in the positive sense.

system (x, y, z) and in the rotated systems (x', y', z') are readily shown to be related by

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} \equiv \mathbf{R}_z(\psi) \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}. \tag{10.13}$$

If this rotated coordinate system is then itself rotated about the y'-axis by an angle θ, as shown in Fig. 10.9, the relation between the coordinates of the point P

in both systems is given by

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix} \begin{pmatrix} x'' \\ y'' \\ z'' \end{pmatrix} \equiv \mathbf{R}_y(\theta) \begin{pmatrix} x'' \\ y'' \\ z'' \end{pmatrix}. \tag{10.14}$$

Combining these two rotations then gives

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \cos\theta\cos\psi & -\sin\psi & \sin\theta\cos\psi \\ \cos\theta\sin\psi & \cos\psi & \sin\theta\sin\psi \\ -\sin\theta & 0 & \cos\theta \end{pmatrix} \begin{pmatrix} x'' \\ y'' \\ z'' \end{pmatrix}. \tag{10.15}$$

The resulting rotation matrix is defined as

$$\mathbf{R}_{zy}(\theta, \psi) = \mathbf{R}_z(\psi)\mathbf{R}_y(\theta) = \begin{pmatrix} \cos\theta\cos\psi & -\sin\psi & \sin\theta\cos\psi \\ \cos\theta\sin\psi & \cos\psi & \sin\theta\sin\psi \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}. \tag{10.16}$$

Rotation matrices have several interesting properties such as the fact that their determinant is always ± 1. More useful is the fact that their inverse $\mathbf{R}^{-1} = \mathbf{R}^{\mathbf{T}}$, i.e., the inverse is the transpose of the matrix.[5]

Now consider a particle moving in direction $\mathbf{\Omega}(u, v, w) = \mathbf{i}\sin\theta\cos\psi + \mathbf{j}\sin\theta\cos\psi + \mathbf{k}\cos\theta \equiv (\sin\theta\cos\psi, \sin\theta\cos\psi, \cos\theta)$ which is scattered through an angle θ_s with a change in azimuth of $\Delta\psi_s$. How does one calculate $\mathbf{\Omega}'(u', v', w')$? First, rotate $\mathbf{\Omega}$ in the laboratory system to a local coordinate system in which the vector lies along the z-axis. It is readily verified that $\mathbf{\Omega}(\theta, \psi) = \mathbf{R}_{zy}(\theta, \psi)(0, 0, 1)$, so it follows that $\mathbf{R}_{zy}^{-1}(\theta, \psi)\mathbf{\Omega} = (0, 0, 1)$.

Next, the transformation $\mathbf{R}_{zy}(\theta_s, \Delta\psi_s)(0, 0, 1)$ produces a vector, in the local coordinate system, with the direction cosines given by the scattering angles, namely, $(\sin\theta_s\cos\Delta\psi_s, \sin\theta_s\sin\Delta\psi_s, \cos\theta_s)$. Finally, this scattered vector is rotated back to the laboratory system with the rotation matrix $\mathbf{R}_{zy}(\theta, \psi)$. Combining these three steps, the direction of the scattered particle is given by

$$\mathbf{\Omega}'(u', v', w') = \mathbf{R}_{zy}(\theta, \psi)\mathbf{R}_{zy}(\theta_s, \Delta\psi_s)\mathbf{R}_{zy}^{-1}(\theta, \psi)\mathbf{\Omega}. \tag{10.17}$$

To relate the new direction cosines (u', v', w') to the before-scattered direction cosines (u, v, w), the rotation matrix of Eq. (10.16) can be written as

$$\mathbf{R}_{zy}(\theta, \psi) = \begin{pmatrix} w\cos\psi & -\sin\psi & u \\ w\sin\psi & \cos\psi & v \\ -\sin\theta & 0 & w \end{pmatrix}. \tag{10.18}$$

[5] Such matrices are often called *orthogonal* matrices.

With this result and the relation $\mathbf{R}_{zy}^{-1}(\theta,\psi)\mathbf{\Omega} = (0,0,1)$, Eq. (10.17) gives the following explicit expressions:

$$u' = u\cos\theta_s + \sin\theta_s(w\cos\psi\cos\Delta\psi_s - \sin\psi\sin\Delta\psi_s), \tag{10.19}$$

$$v' = v\cos\theta_s + \sin\theta_s(w\sin\psi\cos\Delta\psi_s + \cos\psi\sin\Delta\psi_s), \tag{10.20}$$

$$w' = w\cos\theta_s - \sin\theta\sin\theta_s\cos\Delta\psi_s. \tag{10.21}$$

Because $\sin\theta = \sqrt{1-w^2}$, $u = \sin\theta\cos\psi$, and $v = \sin\theta\sin\psi$, these results can be written in an alternative form that minimizes trigonometric function calls, namely,

$$u' = u\mu_s + \sqrt{1-\mu_s^2}(wu\cos\Delta\psi_s - v\sin\Delta\psi_s)/\sqrt{1-w^2}, \tag{10.22}$$

$$v' = v\mu_s + \sqrt{1-\mu_s^2}(wv\cos\Delta\psi_s + u\sin\Delta\psi_s)/\sqrt{1-w^2}, \tag{10.23}$$

$$w' = w\mu_s - \sqrt{1-w^2}\sqrt{1-\mu_s^2}\cos\Delta\psi_s, \tag{10.24}$$

where $\mu_s \equiv \cos\theta_s$.

From Eq. (10.21), explicit expressions for the polar and azimuthal angles after scatter can be obtained, namely,

$$\cos\theta' = \cos\theta_s\cos\theta - \sin\theta_s\sin\theta\cos\Delta\psi_s \tag{10.25}$$

and

$$\psi' = \psi + \cos^{-1}\left[\frac{\cos\theta\cos\theta_s - \cos\theta'}{\sin\theta\sin\theta_s}\right]. \tag{10.26}$$

The above results for calculating (u', v', w') are sufficient for most cases. However, if $\mathbf{\Omega}(u, v, w)$ is almost parallel to the z-axis, then $w \simeq 1$ and the $1/\sqrt{1-w^2}$ term becomes very large and leads to numerical errors. For the case that $w \gtrsim 0.9$, say, an alternative approach is to use a rotation matrix that first rotates about the x-axis followed by a rotation about the y-axis to create a local coordinate system in which $\mathbf{\Omega}$ is pointed along the y-axis. In this case, the rotation matrix is

$$\mathbf{R}_{yx}(\theta,\psi) = \begin{pmatrix} \cos\psi & \sin\theta\sin\psi & \cos\theta\sin\psi \\ 0 & \cos\theta & -\sin\theta \\ -\sin\psi & \sin\theta\cos\psi & \cos\theta\cos\psi \end{pmatrix}. \tag{10.27}$$

Now $\mathbf{R}_{yx}^{-1}(\theta,\psi)\mathbf{\Omega} = (0,1,0)$, and the direction vector, as above, is given by

$$\mathbf{\Omega}' = \mathbf{R}_{yx}(\theta,\psi)\mathbf{R}_{yx}(\theta_s,\Delta\psi_s)\mathbf{R}_{yx}^{-1}(\theta,\psi)\mathbf{\Omega}. \tag{10.28}$$

Explicitly, it is found that, with $\mu_s = \cos\theta_s$,

$$u' = u\mu_s + \sqrt{1-\mu_s^2}(vu\cos\Delta\psi - w\sin\Delta\psi_s)/\sqrt{1-v^2}, \tag{10.29}$$

$$v' = v\mu_s - \sqrt{1-v^2}\sqrt{1-\mu_s^2}\cos\Delta\psi_s, \tag{10.30}$$

$$w' = w\mu_s + \sqrt{1-\mu_s^2}(wv\cos\Delta\psi - u\sin\Delta\psi_s)/\sqrt{1-v^2}. \tag{10.31}$$

In the subsections below, specific approaches for simulating neutron and photon scattering are detailed. For sampling from other reactions that produce secondary particles (e.g., fission, $(n, 2n)$, fluorescence, etc.), the reader is referred to the MCNP manual (MCNP, 2003).

10.6.2 Photon Scattering From a Free Electron

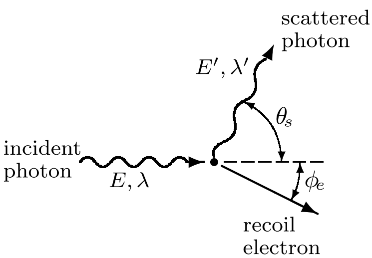


Figure 10.10 A photon with energy E and wavelength λ is scattered by a free electron. After scattering, the photon has a reduced energy E' and larger wavelength λ'.

In Compton scattering, a photon of energy E and dimensionless wavelength $\lambda = m_ec^2/E$ scatters from a free electron as shown in Fig. 10.10. Here m_ec^2 is the rest mass energy equivalent of the electron, namely, 0.511 MeV. The differential scattering cross section is given by the Klein–Nishina formula (Shultis and Faw, 2000). The PDF to sample from this cross section formula can be written in terms of the ratio $r \equiv \lambda'/\lambda = E/E'$ as

$$f(r|\lambda) = \frac{K(\lambda)}{r^2}\left[\frac{1}{r} + r - 1 + (1+\lambda-r\lambda)^2\right], \tag{10.32}$$

where $K(\lambda)$ is the normalization constant to make $f(r|\lambda)$ a PDF (see Problem 10.2). The cosine of the scattering angle is given by

$$\cos\theta_s = 1 + \lambda - r\lambda. \tag{10.33}$$

The range of the scattering angle $0 \le \theta_s \le \pi$ thus gives the constraint $1 \le r \le 1 + (2/\lambda)$. This PDF can be separated into a sum of two PDFs amenable to sampling by the composition-rejection method (see Section 4.1.8) as follows (Kahn, 1954):

$$\begin{aligned} f(r|\lambda) &= K\left\{\left(\frac{1}{r} - \frac{1}{r^2}\right) + \frac{1}{r^2}\left(\frac{1}{r} + [1+\lambda-r\lambda]^2\right)\right\} \\ &= A_1g_1(r)f_1(r) + A_2g_2(r)f_2(r). \end{aligned} \tag{10.34}$$

The constants $A_1 \equiv K/(2\lambda)$ and $A_2 \equiv 4K/(\lambda+2)$ have the properties

$$\frac{A_1}{A_1+A_2} = \frac{\lambda+2}{2+9\lambda} \equiv A_1' \quad \text{and} \quad \frac{A_2}{A_1+A_2} = \frac{8\lambda}{2+9\lambda} \equiv A_2', \tag{10.35}$$

where $A_1' + A_2' = 1$. The functions f_1 and f_2 are defined as

$$f_1(r) \equiv 4\left(\frac{1}{r} - \frac{1}{r^2}\right) \quad \text{and} \quad f_2(r) \equiv \frac{1}{2}\left[\frac{1}{r} + (1+\lambda-r\lambda)^2\right], \tag{10.36}$$

and have the property that $f_i(r) \le 1$ for $1 \le r \le 1 + (2/\lambda)$. Finally, the PDFs $g_1(r) \equiv \lambda/2$ and $g_2(r) \equiv (\lambda+2)/(2r^2)$ have the easily invertible CDFs

$$G_1(r) = \frac{\lambda}{2}(r-1) \quad \text{and} \quad G_2(r) = \frac{\lambda+2}{2}\left(1 - \frac{1}{r}\right). \tag{10.37}$$

Thus Eq. (10.32) can be written as

$$f(r|\lambda) = K'(\lambda)[A_1' g_1(r) f_1(r) + A_2' g_2(r) f_2(r)], \tag{10.38}$$

where the PDF normalization constant $K'(r) = K(r)(9\lambda+2)/[2\lambda(\lambda+2)]$.

The algorithm for sampling from Eq. (10.34) is as described in Section 4.1.8, namely,

1. If $\rho_i < (\lambda+2)/(2+9\lambda)$, calculate $r = 1 + (2\rho_{i+1}+1)/\lambda$ and accept this value if $\rho_{i+2} < (4/r)[1-(1/r)]$.
2. Else, if $\rho_i \ge (\lambda+2)/(2+9\lambda)$, calculate $r = (\lambda+2)/(2\rho_{i+1}+\lambda)$ and accept this value if $\rho_{i+2} < [(1/r) + (1+\lambda-r\lambda)^2]/2$.
3. If no acceptable value for r is found, return to step 1 and try again.

Once a value of r is accepted, the wavelength of the scattered photon is simply $\lambda' = r\lambda$ corresponding to an energy of $E' = E/r$. Lastly, $\cos\theta_s$ is found from Eq. (10.33) and $\sin\theta_s = \sqrt{1-\cos^2\theta_s}$. A comparison of this and other sampling methods for the Klein–Nishina cross section is given by Blomquist and Gelbard (1983).

10.6.3 Neutron Scattering

Neutron scattering data are almost always tabulated in the center-of-mass coordinate system (Shultis and Faw, 2000) in which there is zero net linear momentum before and after the scatter. The microscopic differential cross section for either elastic or inelastic scattering is most often approximated as

$$\sigma_s(E,\omega_c) \simeq \frac{\sigma_s(E)}{4\pi}\sum_{n=0}^{N}(2n+1) f_n(E) P_n(\omega_c), \tag{10.39}$$

in which $P_n(\omega_c)$ is the Legendre polynomial of order n and ω_c is the cosine of the scattering angle in the center-of-mass system. Cross section data consist of tabulating the Legendre coefficients $f_n(E)$ (with f_0 always equal to 1). The reason for this representation is that, in the center-of-mass system, scattering is usually more isotropic than in the laboratory system (target at rest system) so that only a low-order expansion is needed. Indeed, for isotropic scattering $N = 0$, and most data libraries seldom exceed $N = 8$ even for highly anisotropic scattering.[6]

The first step in simulating a neutron scattering event is to construct a PDF with the same shape as Eq. (10.39) and to obtain a sample of ω_c from it. The corresponding cosine of the scattering angle in the laboratory system is (Shultis and Faw, 2000)

$$\mathbf{\Omega}\bullet\mathbf{\Omega}' \equiv \cos\theta_s = \frac{\gamma + \omega_c}{\sqrt{1 + 2\gamma\omega_c + \gamma^2}}, \tag{10.40}$$

where

$$\gamma = \left[A^2 + \frac{A(A+1)Q}{E}\right], \tag{10.41}$$

with Q being the Q-value (Shultis and Faw, 2007) of the scatter ($= 0$ for elastic scatter and < 0 for inelastic scatter) and A being the ratio of the atom's mass to that of the neutron ($\simeq$ atomic mass number). From the conservation of energy and momentum, the energy of the scattered neutrons is given by (Shultis and Faw, 2000)

$$E' = \frac{1}{2}E(1+\alpha) + \frac{1}{2}(1-\alpha)E\omega_c\sqrt{1+\Delta} + \frac{QA}{A+1}, \tag{10.42}$$

in which $\alpha \equiv (A-1)^2/(A+1)^2$ and $\Delta \equiv Q(A+1)/(AE)$.

10.7 Time Dependence

Although most transport problems are independent of time, Monte Carlo simulations are easily extended to time-dependent cases when needed, for example, to estimate when a particle reaches some surface. For neutrons, the energy determines the neutron's speed from which the time for various segments of its history are easily calculated. Tallies are then binned by the time the contribution to the tally was made. Photons are even easier because they all travel at the speed of light. Even time-varying sources can be treated by sampling from an appropriate time distribution to obtain the starting time for each history and tracking the times of the various events that occur during the history. In the following discussion, however, it is assumed that the tallies and flux density are independent of time.

[6] For the case of isotropic scattering (often a good approximation for heavier atoms since the center-of-mass system becomes the target-at-rest system for infinitely heavy scattering centers), the directed cosines of the scattered particle can be calculated in the same manner as used for the directed cosines of an isotropically emitted source particle (see Section 10.3.1).

10.8 Particle Weights

The efficiency of a Monte Carlo transport calculation depends on both the speed with which the calculation can be made and the variance of the result. The speed is, to a large extent, controlled by the hardware and, to a lesser extent, by the computer programmer and the operating system. The variance is controlled primarily by the type of tally used and the number of particles that contribute to the tally. In a strict analog simulation, the only way to reduce the variance of a tally is to run more histories. But this brute-force approach only reduces the variance as $1/N$. As explained in Chapter 5, a better way to increase the precision of the tally is to use nonanalog techniques to force more particles to the regions of phase space where particles are more likely to score without increasing (and perhaps decreasing) the sampling in less important regions of phase space.

In a nonanalog simulation, as explained in Section 4.2.1, samplings for the various physical events used to create a particle history are biased so that events more likely to contribute to the tally are produced more frequently than in an analog simulation. This bias introduced into particle histories must, of course, be corrected when scoring the histories so as to produce an unbiased estimator. For example, if a particular bias makes a history α times more likely to score, then its contribution to the tally must be multiplied by a factor of $1/\alpha$. This removal of biases is readily accomplished by assigning a *weight* W to each particle, and, at every event in its history that introduces a bias, its weight is adjusted so that its score *times* its weight produces an unbiased contribution to the score. For example, if, in a biased sampling for some event, the particle's chance of scoring is changed by a factor of α, its current weight W is modified to W/α. If there is no bias in starting a particle history, the history begins with a weight of unity. Note that weights are not constrained to integer values and may be greater or less than unity.

In short, in a nonanalog Monte Carlo simulation, the scoring of the particles reaching the tally region or detector must be modified from just the sum of the particles' scores to the sum of the particles' weighted scores. In this way, unbiased results can be achieved with less computing time, if proper variance reduction methods are used, compared to an analog calculation producing the same results and precision.

10.9 Scoring and Tallies

After a particle history ends or, more usually, after a particle leaves a cell, its contribution to the score (or tally) is added to the running score of interest. Almost anything of interest that depends on the radiation field can be estimated from the particle histories used in a Monte Carlo simulation. Here, some of the more common tallies are reviewed. For more thorough treatments and other estimators, the reader is referred to Kalos et al. (1968), Carter and Cashwell (1975), and MCNP (2003).

10.9.1 Fluence Averaged Over a Surface

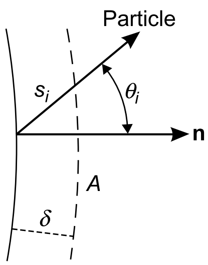


Figure 10.11 Particle crossing a surface at an angle θ_i from the outward normal $\mathbf{n}$.

Often the fluence, averaged over some surface (or portion of a surface), is sought. Imagine a parallel surface, a very small distance δ from the surface of interest as shown in Fig. 10.11. From Eq. (9.13), the fluence (flux density integrated over all time or all simulated histories) is just the weighted sum of the path lengths of all the particles passing through this incremental volume $\Delta V = A\delta$ divided by the volume of the extended region, i.e., $\Phi = \sum_i W_i s_i / \Delta V$, where W_i is the weight of particle i. The path length for the particle shown is $s_i = \delta/|\cos\theta_i|$, where θ_i is the angle between the particle's exit direction and the outward normal $\mathbf{n}$. Thus, the fluence contribution of the ith particle crossing the surface is $\Phi_i = \lim_{\delta\to 0}(W_i\delta/|\cos\theta_i|)/A\delta = W_i/(A|\cos\theta_i|) = W_i/(A|\mathbf{\Omega}_i \bullet \mathbf{n}|)$. Every time a particle crosses the surface, the value of its $W_i/\cos\theta_i$ is added to the tally. Of course, many histories may not cross the surface in question, so no score is contributed to the tally. Finally, after N histories the average fluence, *per source particle*, is estimated as

$$\overline{\Phi}_S \equiv \frac{1}{A}\int_A dA \int_0^\infty dE \int_{4\pi} d\mathbf{\Omega}\, \Phi(\mathbf{r}, E, \mathbf{\Omega}) \simeq \frac{1}{NA}\sum_{i=1}^{N}\sum_{j=1}^{n_i} \frac{W_i^j}{|\cos\theta_i^j|}, \qquad (10.43)$$

where n_i is the number of surface crossing made by the ith particle,[7] W_i^j is the ith particle's weight on its jth crossing, and θ_i^j is the angle with respect to the outward normal at the jth crossing. Here $\Phi(\mathbf{r}, E, \mathbf{\Omega})$ is the energy and angular fluence normalized to one source particle.

It should be noted that the variance of this fluence estimator is infinite and, hence, the central limit theorem cannot be used to estimate confidence intervals. If many particles cross the surface in nearly tangential directions, i.e., $\cos\theta$ is very small, the tally contributions become very large. To avoid such large scoring events, many Monte Carlo transport codes use an ad hoc fixup procedure to avoid large variances. If $|\cos\theta| < \epsilon$, where ϵ is some small value, say 0.01, then replace the $|1/\cos\theta_i^j|$ in Eq. (10.43) by $2/\epsilon$, the expected value of $1/|\cos\theta|$ when $|\cos\theta| < \epsilon$ and the fluence is nearly isotropic. Although this procedure is not rigorous, in most cases reliable estimates of the surface-averaged fluence are obtained.

[7] Typically n_i is 0 or 1. But a particle may make several crossing, especially if the surface is re-entrant.

If the angular distribution and energy spectra of the particles crossing the surface are of interest, then the tally procedure requires the use of a multi-dimensional array for scoring. Polar angle cosines, $\cos\theta$, can be divided into some number of intervals from -1 to 1. Likewise, the azimuthal angle about the outward normal to the surface can be divided into some number of intervals from 0 to 2π radians. Finally, the full range of possible particle energies may also be divided into some number of intervals. When a particle is found to cross the surface, a score is made in the *bin* associated with the energy and direction of the particle. Binning by energy also allows each bin to be multiplied by an energy-dependent response function $\mathcal{R}(E)$, so that, for example, summing over all energy and angular bins, the average dose at the surface can be estimated. In a similar fashion, multiplication by a reaction interaction coefficient $\mu_j(E)$ and summation of the bin results yields the average reaction density $\int \mu_j(E)\overline{\Phi}(E)\,dE$ at the surface.

10.9.2 Fluence in a Volume: Path Length Estimator

The idea behind Eq. (9.13) can also be extended to determine the average fluence in the volume of a cell. In this type of tally, the sum of path lengths s_i each particle makes in the volume V of interest is accumulated (see Fig. 10.12). Again, if a particle history never reaches the cell in question, then $s_i = 0$. The average fluence, per source particle, is then estimated as

$$\overline{\Phi}_V \simeq \frac{1}{NV}\sum_{i=1}^{N}\sum_{j=1}^{n_i} W_i^j s_i^j, \qquad (10.44)$$

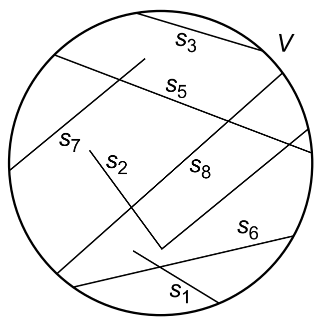


Figure 10.12 Particle paths in a cell of volume V.

where n_i is the number of times the ith particle enters V, s_i^j is the jth track length in V, and W_i^j is the particle's weight when entering V for the jth time. By breaking the energy range into a contiguous set of bins and adding each weighted path length to the tally in the appropriate energy bin, the energy spectrum of the volume-averaged fluence can be estimated as

$$\overline{\Phi}_V(E) \equiv \frac{1}{V}\int_V dV \int_{4\pi} d\mathbf{\Omega}\,\Phi(\mathbf{r}, E, \mathbf{\Omega}) \simeq \frac{S_i}{NV\Delta E}, \qquad (10.45)$$

where ΔE is the width of the energy bin containing E and S is the weighted sum of path lengths tallied in that energy bin. As before the tally for each particle can be multiplied by some function $\mathcal{R}(E)$ so that when the results in each energy bin are summed, the average value of $\int dE\,\mathcal{R}(E)\overline{\Phi}_V(E)$ is estimated.

If the volume V is large so that many histories enter it, the path length estimator is generally a good tally. If the region is a thin curved shell, most track lengths have similar lengths and the estimator has a small variance. By contrast, for a region bounded by two closely spaced parallel planes, there may be a wide variation in the track lengths through the region, and the resulting path length estimator may have a large variance. The path length estimator is also computationally quite efficient, since particle tracking already computed the track lengths in the regions, so little extra effort is needed for this tally.

10.9.3 Fluence in a Volume: Reaction Density Estimator

An alternative estimator for the average fluence in a region is obtained by tallying the number of interactions that occur in the region. From Eq. (9.37), the average density of reactions in a homogeneous volume V is found to be

$$\overline{R}(E) \equiv \frac{1}{V}\int_V \mu_t(\mathbf{r}, E)\Phi(\mathbf{r}, E)dV = \frac{\mu_t(E)}{V}\int_V \Phi(\mathbf{r}, E)\,dV \equiv \mu_t(E)\overline{\Phi}_V(E). \tag{10.46}$$

Thus, the contribution to the average fluence in V made by a particle whose energy E lies in ΔE and that collides in V is made by adding $1/\mu_t(E)$ to the tally energy bin. Thus, after N histories, the estimate of the average fluence is

$$\overline{\Phi}_V(E) \equiv \frac{1}{V}\int_V dV \int_{4\pi} d\mathbf{\Omega}\,\Phi(\mathbf{r}, E, \mathbf{\Omega}) \simeq \frac{1}{NV\Delta E}\sum_{i=1}^{N}\sum_{j=1}^{n_i}\frac{W_i^j}{\mu_t(E)}, \tag{10.47}$$

where n_i is the number of collisions in V made by particle i when it had an energy in ΔE and the superscript j refers to the jth such interaction in V made by the ith particle.

The probability a particle interacts while crossing a cell is

$$p = 1 - \exp[-\mu_t(E)s_{\max}],$$

where $s_{\max}$ is the maximum distance the particle could travel in the cell along its flight path. For thin or small regions, this probability is small and the above estimator will produce poor results. A more efficient way to estimate the fluence is to tally the interaction contribution times the probability the particle interacts, i.e.,

$$\overline{\Phi}_V(E) \simeq \frac{1}{NV\Delta E}\sum_{i=1}^{N}\sum_{j=1}^{n_i} W_i^j \frac{1-e^{-\mu_t(E)s_{\max,j}}}{\mu_t(E)}, \tag{10.48}$$

where n_i now is the number of cell crossings made by particle i when it had energy E in ΔE.

For the reaction density estimator of Eq. (10.47) to work properly, particles must cross the cell of interest. If the cell is far from the source, even if a particle heads toward the cell, it is unlikely to reach it and contribute to the tally. But all is not lost. The contribution to the fluence in the cell, however, can be calculated for particles heading toward the cell but that interact at some point before reaching it. Denote the distance from the interaction point to the nearest cell surface by d along the original trajectory. One simply includes in the estimator of Eq. (10.47) a term that gives the probability p_{hit} that the particle *does* reach the cell,

$$p_{hit} = \exp\left[-\int_0^d \mu_t(\mathbf{r}, E)ds\right],$$

where the integration is along the trajectory. Thus, the argument of the exponential is just the number of mean free path lengths the particle would have to travel from its interaction point to reach the cell. The average fluence in the cell is then estimated as

$$\overline{\Phi}_V(E) \simeq \frac{1}{NV\Delta E}\sum_{i=1}^{N}\sum_{j=1}^{n_i} W_i^j \frac{1-e^{-\mu_t(E)s_{\max,j}}}{\mu_t(E)} \exp\left[-\int_0^{d_j} \mu_t(\mathbf{r}, E)ds\right]. \tag{10.49}$$

Here n_i is the number of collision, both inside and outside the cell, made by particle i. If a collision is inside the cell, then, of course, $d_j = 0$.

For a large thin region, subtending a large solid angle, Eq. (10.49) may be a good estimator; but for small distant regions few particles will even interact while heading toward it. Even this can be overcome by including the probability that, if the collision was a scatter, the particle would have an energy in ΔE and head toward the cell. This idea of forcing a scatter toward a cell of interest is developed further in Section 10.9.5.

10.9.4 Average Current Through a Surface

Sometimes the average total flow or current through a surface is sought (see Section 9.1.5). For particles with energy E in ΔE and direction $\mathbf{\Omega}$ in $\Delta\Omega$, the

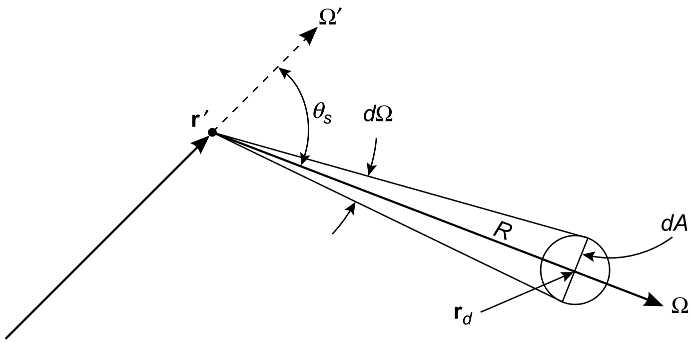


Figure 10.13 A particle moving in direction $\mathbf{\Omega}'$ makes a collision at $\mathbf{r}'$ and scatters toward a small spherical detector at $\mathbf{r}_d$ of cross-sectional area dA and a distance $R = |\mathbf{r}' - \mathbf{r}_d|$ from the scattering point.

estimate of the average current, per source particle, is[8]

$$\overline{J}(E, \mathbf{\Omega}) \equiv \frac{1}{A}\int_A dA \int_{\Delta\mathbf{\Omega}} d\mathbf{\Omega}\, |\mathbf{\Omega}\bullet\mathbf{n}|\Phi(\mathbf{r}, E, \mathbf{\Omega}) \simeq \frac{1}{NA\,\Delta E\Delta\Omega}\sum_{i=1}^{N} n_i W_i, \quad (10.50)$$

where n_i is number of crossings of the surface made by particle i when it had energy in ΔE and direction in $\Delta\Omega$. Typically, n_i is 0 or 1, but a particle may cross a re-entrant surface several times.

If the net flows across a surface are wanted (see Eq. (9.20)), then use the above procedure with only the two cosine ($\mu \equiv \mathbf{n}\bullet\mathbf{\Omega}$) angular bins $-1 \le \mu < 0$ and $0 < \mu \le 1$ to estimate $J^-(E_i)$ and $J^+(E_i)$, respectively. The net flow is $J_{net} = J^+ - J^-$.

10.9.5 Fluence at a Point: Next-Event Estimator

A very powerful technique for scoring is to combine deterministic tally contributions and the stochastic collisions that occur during a history. One such estimator is the *next-event estimator*, also known as the no-further-collision estimator. Consider the small spherical detector, with cross-sectional area dA, shown in Fig. 10.13. A particle traveling in direction $\mathbf{\Omega}'$ has a collision at $\mathbf{r}'$. The collision may not be a scatter, and even if it was a scatter, it is very unlikely to scatter in the direction of dV or even reach it to contribute to the fluence tally for dV. However, one can analytically calculate the probability the particle will scatter at $\mathbf{r}'$ and reach the detector without further interaction, and thus provide a contribution to the fluence tally for the detector, thereby short-cutting the Monte Carlo process.

[8] Here, J is the time-integrated current vector discussed in Section 9.1.4 just as the fluence Φ is the time-integrated flux ϕ (see Eq. (9.15)).

The probability the particle with energy E scatters at $\mathbf{r}'$ through a scattering angle θ_s and has a new direction of travel in $d\Omega$ about $\boldsymbol{\Omega}$ with energy E' is

$$p_1 = \frac{\mu_s(\mathbf{r}', E, \omega_s)\, d\Omega}{\int_0^{2\pi} d\psi \int_{-1}^{1} d\omega_s \mu_s(\mathbf{r}', E, \omega_s)} = \frac{1}{2\pi}\left[\frac{\mu_s(\mathbf{r}', E, \omega_s)}{\int_{-1}^{1} d\omega_s \mu_s(\mathbf{r}', E, \omega_s)}\right] d\Omega$$
$$= \frac{p(\omega_s)}{2\pi} d\Omega,$$

where $p(\omega_s)$ is the PDF (defined by the terms in the square brackets) for scattering through an angle $\theta_s = \cos^{-1} \boldsymbol{\Omega}\bullet\boldsymbol{\Omega}'$. The distance between $\mathbf{r}'$ and $\mathbf{r}_d$ is $R = |\mathbf{r}' - \mathbf{r}_d|$ so that $d\Omega = dA/R^2$. The probability that such a scattered particle actually reaches the detector without further collision is

$$p_2 = \exp\left[-\int_0^R \mu_t(\mathbf{r}, E')\, ds\right],$$

where the argument of the exponential is the total number of mean free path lengths between $\mathbf{r}'$ and $\mathbf{r}_d$.

As stated by Eq. (9.14), the flux density can be interpreted as the number of particles entering a spherical detector of cross-sectional area dA divided by dA. Thus, the contribution to the fluence made by the particle with weight W that interacts at $\mathbf{r}'$ is

$$\delta\Phi = W \frac{\mu_s(\mathbf{r}', E)}{\mu_s(\mathbf{r}', E')} \frac{p_1 p_2}{dA} = W \frac{\mu_s(\mathbf{r}', E)}{\mu_s(\mathbf{r}', E')} \frac{p(\omega_s)}{2\pi R^2} \exp\left[-\int_0^R \mu_t(\mathbf{r}, E)\, ds\right]. \quad (10.51)$$

This result is independent of dA and, thus, is used as a fluence tally for a point detector at $\mathbf{r}_d$. After each interaction in a particle's history, an estimate is made deterministically for the expected contribution of that interaction to the fluence at point $\mathbf{r}_d$.

A difficulty with this point detector tally occurs when the tally point lies within a scattering medium. Because of the $1/R^2$ term in Eq. (10.51), an enormous contribution is made to the tally if an interaction occurs very near the tally point. In fact, in a scattering region the variance of this estimator is infinite! An infinite variance, however, does not mean the tally cannot be used; indeed the tally will converge, but the asymptotic behavior is reduced to $1/\sqrt[3]{N}$, slower than the $1/\sqrt{N}$ convergence for a tally with a finite variance (Kalos et al., 1968). Of course, if the tally point is in a vacuum or nonscattering medium no such problem occurs.

The way to avoid an infinite variance in a scattering medium is to surround the tally point by a small sphere of radius R_o, and do no estimation if an interaction occurs at $R < R_o$. Alternatively, if an interaction occurs with $R < R_o$, tally the average fluence uniformly distributed in the volume, i.e.,

$$\delta\Phi(R < R_o) = \frac{\int \Phi\, dV}{\int dV} = W \frac{\mu_s(\mathbf{r}', E)}{\mu_s(\mathbf{r}', E')} \frac{p(\omega_s)}{2\pi} \frac{\int_0^{R_o} \frac{1}{r^2} e^{-\mu_t(E')r}(4\pi r^2)\, dr}{\frac{4}{3}\pi R_0^3}$$

$$= W \frac{\mu_s(\mathbf{r}', E)}{\mu_s(\mathbf{r}', E')} p(\omega_s) \frac{3(1 - e^{-\mu_t(E')R_o})}{2\pi R_o^3}. \qquad (10.52)$$

Although the above next-event estimator can produce good prediction of the fluence at a point, its convergence may be rather slow. This estimator needs many statistical tests to ensure that sound results are achieved. Lux and Koblinger (1991) and MCNP (2003) discuss variance issues associated with the use of next-event estimators to determine point values of the fluence and related quantities.

10.9.6 Flow Through a Surface: Leakage Estimator

Another partially deterministic method is the *leakage estimator* used to find the probability P_{esc} that a particle entering a region leaves through some surface of the region. The idea is very similar to the next-event estimator. At the point of entry of a particle into the region (either entering from outside or born in the region), and after each scattering interaction in the region, a determination is made of the probability that, without further interaction, the photon can escape from the slab. That probability is then scored as a contribution to the flow across the surfaces.

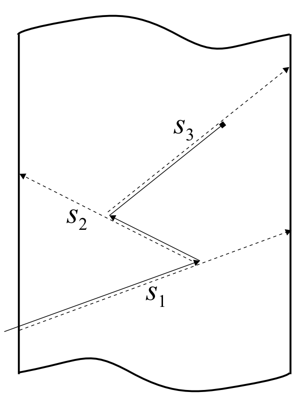


Figure 10.14 A three-segment particle path ending in an absorption.

Thus, for the three-segment path ending in an absorption shown in Fig. 10.14, the leakage tally for this particle history is

$$e^{-\mu s_1} + e^{-\mu s_2} + e^{-\mu s_3}.$$

Also note that if the particle passes through the surface on its last segment in the region, no score is tallied. This method can also be applied to scoring the energy spectrum and angular distribution of the flow rates through each slab face. For deep penetration problems, score contributions may vary considerably among histories, so that large variances result. An example of an analog leakage calculation and one using the leakage estimator is given in Example 10.2.

Example 10.2 Escape Probability From an Infinite Slab

Consider an infinite homogeneous slab surrounded by a vacuum. One-speed particles are born uniformly throughout the slab. The particles scatter isotropically with $c = \mu_s/\mu$, the scattering probability per interaction. The problem is to calculate the probability P_{esc} that a particle born in the slab eventually escapes of leaks from it.

In a straight analog simulation of N histories in which n_{esc} particles leak from the slab,

$$p_{esc} = \frac{n_{esc}}{N}.$$

However, for thick slabs with small values of c only particles born near the surface have much chance of escape and of contributing to the tally. With a leakage estimator, *every* particle contributes something to the tally, although many contributions are quite small. In Fig. 10.15 results are shown for the case $c = 0.1$ using analog scoring (left-hand figure) and the leakage estimator (right figure). It is seen that, although both methods give comparable results for thin slabs, the leakage estimator is better for the thick slabs.

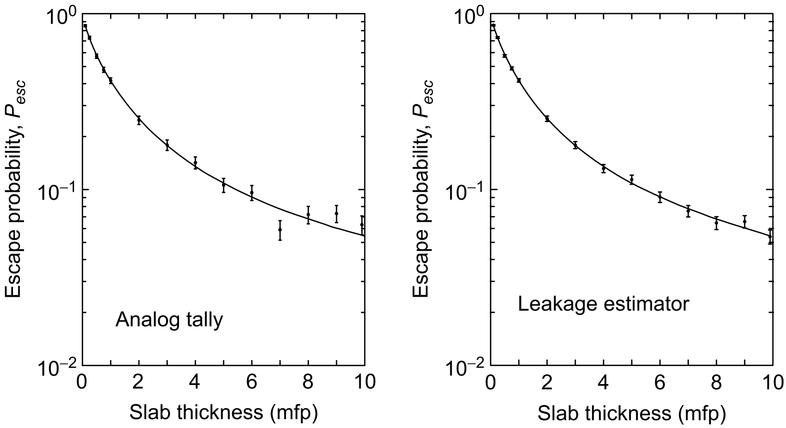
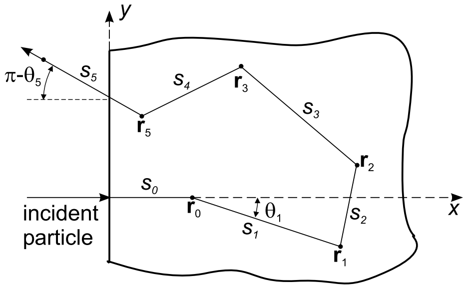


Figure 10.15 Escape probability for particles born uniformly in an infinite homogeneous slab with $c = 0.1$. Monte Carlo results with $\pm\sigma$ error bars are for 1000 particles. Results on the left are for an analog tally, i.e., P_{esc} = (no. escaping)/(no. born). Results on the right were obtained using the leakage estimator. The solid lines on these plots are cubic spline fits to results obtained with 10^7 histories and that contain negligible error.

10.10 An Example of One-Speed Particle Transport

Before venturing into more detailed aspects of Monte Carlo transport theory, it might be useful to consider a simple example that uses many of the concepts introduced up to now. In Example 10.3, a homogeneous half-space is normally and uniformly illuminated by incident particles. The problem is to find the distribution of reflected particles.

Example 10.3 Particle Reflection From a Half-Space



The challenge for solving the transport equation occurs when particles can scatter as well as be absorbed. Analytic solutions are known only for highly simplified forms of the transport equation Eq. (9.50), and then only with substantial mathematical effort. To allow a comparison between a Monte Carlo calculation and an exact analytical solution, the following simplifications are applied to Eq. (9.50): (1) particles are assumed to have only a single energy, (2) scattering is isotropic, (3) the medium is a source-free homogeneous half-space $x \geq 0$, and (4) particles are incident uniformly and normally on the surface of the half-space. Under these conditions, Eq. (9.50) can be written as

$$\omega \frac{\partial \phi(x,\omega)}{\partial x} + \phi(x,\omega) = \frac{c}{2}\int_{-1}^{1} \phi(x,\omega')\, d\omega', \qquad 0 \leq x < \infty, \quad -1 \leq \omega \leq 1, \quad (10.53)$$

where x is measured in mean free path lengths, $\omega = \mathbf{i} \bullet \boldsymbol{\Omega}$, and $c = \mu_s/\mu \leq 1$. The boundary conditions of this *albedo problem* are (1) $\phi(0,\omega) = \delta(\omega - 1)$ and (2) $\lim_{x\to\infty} \phi(x,\omega) < \infty$.[9] In this formulation, the units of $\phi(x,\omega)$ are no longer $\text{cm}^{-2}\,\text{s}^{-1}\,\text{sr}^{-1}$ but $\text{cm}^{-2}\,\text{s}^{-1}$ per unit ω, since, to obtain Eq. (10.53), integration over all azimuthal angles has been done.

Several different (and complex) analytical methods can be used to obtain equivalent closed-form analytical expressions for the angular flux density $\phi(x,\omega)$. In particular, the reflected flux density is (Chandrasekhar, 1960)

$$\phi(0,\omega) = \frac{c}{2}\frac{1}{1+|\omega|} H(c,|\omega|) H(c,1), \qquad -1 \leq \omega \leq 0, \qquad (10.54)$$

where $H(c, \cos\theta)$ is Chandrasekhar's H-function, which can be evaluated by an iterative solution of the nonlinear integral equation (Shultis and Faw, 2000)

$$\frac{1}{H(c,\cos\theta)} = \sqrt{1-c} + \frac{c}{2}\int_0^1 \frac{uH(c,u)}{u+\cos\theta}\, du. \qquad (10.55)$$

[9] This albedo problem is used, for example, to describe light reflected from a planetary atmosphere or thermal neutrons reflected from the surface of a shield.

Not only is the analysis to obtain Eq. (10.54) difficult, so is its numerical evaluation, which requires a numeral solution of Eq. (10.55). By contrast, the reflected flux density is easily calculated by a Monte Carlo simulation if the problem is first transformed to an easier equivalent one, as discussed in Section 5.1. Rather than having particles incident all over the infinite surface of the half-space and scoring only those that flow back through a given unit area on the surface after scattering in the half-space, it is far better to have all incident particles impinge on the same point on the surface (say, $y = z = 0$) and to score particles that emerge anywhere on the surface. Because the half-space is infinite in the y- and z-directions, these coordinates have no effect on whether a particles escape, and, hence, only the x-coordinate of a particle's interaction site needs to be calculated.

The Monte Carlo simulation proceeds as follows. An incident particle penetrates a distance s_0, randomly sampled from $f(s) = \mu e^{-s}$, where it is scattered, with probability c, or absorbed, with probability $1 - c$. If a scatter occurs a new direction θ_0 is selected, where $\cos\theta_0$ is sampled from $\mathcal{U}(-1, 1)$, and another path length is randomly chosen to the next interaction point $\mathbf{r}_1$. This procedure is continued, as shown in the figure on the previous page, until the particle either is absorbed or escapes through the surface. The escaping particle is then binned according to its angle of escape.

One possible algorithm for this albedo simulation is given below. Here each ρ refers to a new random number from $\mathcal{U}(0, 1)$.

1. Set up and initial to zero, an M-component vector $\mathbf{g}$ in which g_m accumulates reflected angular flux estimates of $\phi(0, -|\omega|)$ with $0 < \omega_m < |\omega| \le \omega_{m+1} \le 1$. Note $\omega_1 = 0$ and $\omega_{M+1} = 1$, so bin g_k accumulates flux tallies for reflected particles whose $|\omega|$ is in the range $(\omega_k, \omega_{k+1}]$.
2. Generate the first collision site $x_0 = s_0 = -\ln\rho_0$. Set $i = 0$.
3. If $\rho > c$ the particle is absorbed, so go to step 1; else:
   (a) Pick a new cosine direction $\omega_{i+1} = 2\rho - 1$.
   (b) Pick the distance to the next interaction site as $s_{i+1} = -\ln\rho$.
   (c) Calculate the x-coordinate of the interaction site as $x_{i+1} = x_i + \omega_{i+1}s_{i+1}$.
   (d) If $x_{i+1} \le 0$, the particle escapes through the illuminated surface, i.e., it is reflected. Note ω_{i+1} must be negative to escape the half-space. Bin the flux density estimator $1/|\omega_{i+1}|$ in the reflection tally vector $g_k = g_k + 1/|\omega_{i+1}|$ if $\omega_k < \omega_{i+1} \le \omega_{k+1}$. Time for another history, so go to step 2.
   (e) Else, because $0 < x_{i+1} < \infty$, the particle is still in the half-space. If $\rho < c$, the particle scatters so set $i = i + 1$ and go to step 3a; otherwise the particle is absorbed so go to step 2.

After a large number of histories N, the estimate of the reflected angular flux density $\phi(0, \omega)$, $\omega \le 0$, is simply $g(\Delta\omega_i)/N$. In Fig. 10.16, a comparison is shown between the analytical result of Eq. (10.54) and the above Monte Carlo procedure. To obtain such good agreement, 10^7 histories were used.

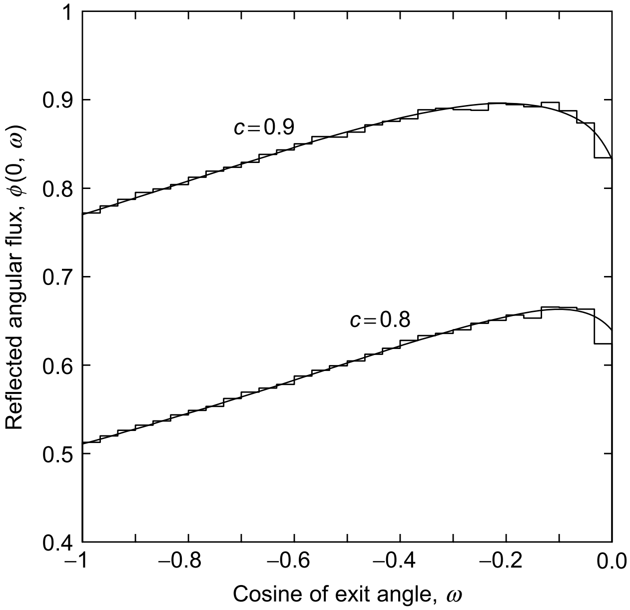


Figure 10.16 Analytical (smooth lines) and Monte Carlo (histograms) results for the albedo problem. The Monte Carlo results used 10^7 histories and 30 cosine bins.

10.11 Monte Carlo Based on the Integral Transport Equation

As discussed in earlier chapters, Monte Carlo is ideally suited for evaluating multidimensional integrals (see Section 2.6) and for solving integral equations (see Section 8.2.4). Thus, it is not surprising that early transport calculations used Monte Carlo techniques on the integral form of the transport equation (Section 9.4).

10.11.1 The Integral Transport Equation

One widely used way to develop Monte Carlo calculational procedures for particle transport is to begin with some form of the integral transport equation (see Section 9.4). It does not really matter which form of the integral equation is used, but here the procedure of Carter and Cashwell (1975) is used, which is based on the integral equation for the collision rate density $F(\mathbf{r}, E, \mathbf{\Omega})$ that is given by Eq. (9.76), namely,

$$F(\boldsymbol{P}) = \int_R d\boldsymbol{P}'\overline{K}(\boldsymbol{P}' \to \boldsymbol{P})F(\boldsymbol{P}') + \overline{S}(\boldsymbol{P}). \tag{10.56}$$

Here $\boldsymbol{P}$ are the particle phase space coordinates $(\mathbf{r}, E, \mathbf{\Omega})$ and region R of phase space contains the volume V of the ambient medium, the energy range of

E, and the 4π sr of directions; $\overline{S}(\boldsymbol{P})$ is the first-flight collision rate density of particles streaming from the source and is given by

$$\overline{S}(\boldsymbol{P}) \equiv \mu(\mathbf{r}, E) \int_V dV' \exp[-\tau(E, \mathbf{r}, \mathbf{r}')]\, \delta\left(\boldsymbol{\Omega} - \frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|}\right) \frac{S(\mathbf{r}', E, \boldsymbol{\Omega})}{|\mathbf{r}-\mathbf{r}'|^2}$$

and $\overline{K}(\boldsymbol{P} \to \boldsymbol{P})$ is the next-flight collision rate density at $\boldsymbol{P}$ due to a collision at $\boldsymbol{P'}$, given by

$$\overline{K}(\boldsymbol{P'} \to \boldsymbol{P}) \equiv \frac{\mu(\mathbf{r}, E)}{\mu(\mathbf{r}', E')} \frac{\mu(\mathbf{r}', E' \to E, \boldsymbol{\Omega}' \to \boldsymbol{\Omega})}{|\mathbf{r}-\mathbf{r}'|^2} \exp[-\tau(E, \mathbf{r}, \mathbf{r}')]\, \delta\left(\boldsymbol{\Omega} - \frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|}\right).$$

Further, suppose the purpose of the Monte Carlo calculation is to estimate the value of some weighted average of $F(\boldsymbol{P})$, namely,

$$G = \int d\boldsymbol{P}\, g(\boldsymbol{P}) F(\boldsymbol{P}), \tag{10.57}$$

where $g(\boldsymbol{P})$ is a prescribed weight function, such as a detector response function, that gives the contribution to G from a collision at $\boldsymbol{P}$.

Formal Solution of the Integral Transport Equation

The integral equation Eq. (10.56) can be solved using the Neumann iteration scheme of Eq. (8.53), namely,

$$F_n(\boldsymbol{P}) = \int d\boldsymbol{P}' \overline{K}(\boldsymbol{P'} \to \boldsymbol{P}) F_{n-1}(\boldsymbol{P'}) + \overline{S}(\boldsymbol{P}), \quad n = 1, 2, 3, \ldots, \tag{10.58}$$

where $F_0(\boldsymbol{P}) = 0$. Thus

$$\begin{aligned}
F_1(\boldsymbol{P}) &= \overline{S}(\boldsymbol{P}), \\
F_2(\boldsymbol{P}) &= \overline{S}(\boldsymbol{P}) + \int d\boldsymbol{P}_1 \overline{K}(\boldsymbol{P}_1 \to \boldsymbol{P}) \overline{S}(\boldsymbol{P}_1), \\
F_3(\boldsymbol{P}) &= \overline{S}(\boldsymbol{P}) + \int d\boldsymbol{P}_2 \overline{K}(\boldsymbol{P}_2 \to \boldsymbol{P}) F_2(\boldsymbol{P}_2) \\
&= \overline{S}(\boldsymbol{P}) + \int d\boldsymbol{P}_1 \overline{K}(\boldsymbol{P}_1 \to \boldsymbol{P}) \overline{S}(\boldsymbol{P}_1) \\
&\quad + \int d\boldsymbol{P}_1 \int d\boldsymbol{P}_2 \overline{K}(\boldsymbol{P}_1 \to \boldsymbol{P}_2) \overline{K}(\boldsymbol{P}_2 \to \boldsymbol{P}) \overline{S}(\boldsymbol{P}_1),
\end{aligned}$$

or, in general,

$$F_n(\boldsymbol{P}) = \overline{S}(\boldsymbol{P}) + \sum_{m=1}^{n-1} \int d\boldsymbol{P}_1 \ldots \int d\boldsymbol{P}_m \overline{K}(\boldsymbol{P}_1 \to \boldsymbol{P}_2) \ldots \overline{K}(\boldsymbol{P}_m \to \boldsymbol{P}) \overline{S}(\boldsymbol{P}_1).$$

If the series converges, the solution of Eq. (10.56) is $F(\boldsymbol{P}) = \lim_{n\to\infty} F_n(\boldsymbol{P})$, i.e.,

$$F(\boldsymbol{P}) = \overline{S}(\boldsymbol{P}) + \sum_{m=1}^{\infty} \int d\boldsymbol{P}_1 \ldots \int d\boldsymbol{P}_m \overline{K}(\boldsymbol{P}_1 \to \boldsymbol{P}_2) \ldots \overline{K}(\boldsymbol{P}_m \to \boldsymbol{P}) \overline{S}(\boldsymbol{P}_1). \tag{10.59}$$

Formal Evaluation of G

The value of G can now be formally obtained by substitution of the collision rate density given by Eq. (10.59) into Eq. (10.57). The result is

$$G = \int d\boldsymbol{P}_1 \overline{S}(\boldsymbol{P}) + \sum_{m=1}^{\infty} \int d\boldsymbol{P}_1 \ldots \int d\boldsymbol{P}_{m+1} \times$$
$$\overline{S}(\boldsymbol{P}_1)\overline{K}(\boldsymbol{P}_1 \to \boldsymbol{P}_2) \ldots \overline{K}(\boldsymbol{P}_m \to \boldsymbol{P}_{m+1}) \overline{S}(\boldsymbol{P}_1) g(\boldsymbol{P}_{m+1}),$$

which can be written more succinctly as

$$G = \sum_{m=0}^{\infty} \int d\boldsymbol{P}_1 \ldots \int d\boldsymbol{P}_{m+1} \overline{S}(\boldsymbol{P}_1)\overline{K}(\boldsymbol{P}_1 \to \boldsymbol{P}_2) \ldots \overline{K}(\boldsymbol{P}_m \to \boldsymbol{P}_{m+1}) \overline{S}(\boldsymbol{P}_1) g(\boldsymbol{P}_{m+1}). \tag{10.60}$$

Evaluation of G by Monte Carlo

To evaluate Eq. (10.60) by Monte Carlo methods, the kernel $\overline{K}(\boldsymbol{P}' \to \boldsymbol{P})$ is first decomposed into a product of a nonabsorption probability, a normalization factor to account for multiplying interactions such as fission or $(n, 2n)$ reactions, and a normalized kernel, namely,

$$\overline{K}(\boldsymbol{P}' \to \boldsymbol{P}) = [1 - \alpha(\boldsymbol{P}')]\eta(\boldsymbol{P}')\kappa(\boldsymbol{P}' \to \boldsymbol{P}). \tag{10.61}$$

Here $\alpha(\boldsymbol{P}') = \mu_a(\boldsymbol{P}')/\mu(\boldsymbol{P}')$ is the absorption (capture) probability at $\boldsymbol{P}'$[10] and $\eta(\boldsymbol{P}')$ is the normalization factor

$$\eta(\boldsymbol{P}') = \frac{\int d\boldsymbol{P}\, \overline{K}(\boldsymbol{P}' \to \boldsymbol{P})}{1 - \alpha(\boldsymbol{P}')}. \tag{10.62}$$

The normalized collision kernel $\kappa(\boldsymbol{P}' \to \boldsymbol{P})$ is

$$\kappa(\boldsymbol{P}' \to \boldsymbol{P}) = \frac{\overline{K}(\boldsymbol{P}' \to \boldsymbol{P})}{\int d\boldsymbol{P}''\, \overline{K}(\boldsymbol{P}' \to \boldsymbol{P}'')}, \tag{10.63}$$

[10] If the spatial coordinates of $\boldsymbol{P}'$ lie outside V, the particle has leaked from the system and $\alpha(\boldsymbol{P}')$ is set to unity, the same as if the particle had been absorbed.

which has the property that $\int d\boldsymbol{P}\,\kappa(\boldsymbol{P}'\to\boldsymbol{P}) = 1$ so that $\kappa(\boldsymbol{P}'\to\boldsymbol{P})$ is a conditional PDF from which the next collision point $\boldsymbol{P}$ can be sampled given the previous nonabsorption collision was at $\boldsymbol{P}'$.

Now apply the factorization of Eq. (10.61) to Eq. (10.60) to obtain

$$G = \sum_{m=0}^{\infty}\int d\boldsymbol{P}_1\int d\boldsymbol{P}_2\ldots\int d\boldsymbol{P}_{m+1}\overline{S}(\boldsymbol{P}_1)[1-\alpha(\boldsymbol{P}_0)]\kappa(\boldsymbol{P}_1\to\boldsymbol{P}_2)\times$$
$$[1-\alpha(\boldsymbol{P}_2)]\kappa(\boldsymbol{P}_2\to\boldsymbol{P}_3)\ldots[1-\alpha(\boldsymbol{P}_m)]\kappa(\boldsymbol{P}_m\to\boldsymbol{P}_{m+1})\times$$
$$\alpha(\boldsymbol{P}_{m+1})W(\boldsymbol{P}_0,\boldsymbol{P}_1,\ldots,\boldsymbol{P}_{m+1}), \tag{10.64}$$

where W is defined as

$$W(\boldsymbol{P}_1,\boldsymbol{P}_2,\ldots,\boldsymbol{P}_{m+1}) \equiv \frac{g(\boldsymbol{P}_{m+1})}{\alpha(\boldsymbol{P}_{m+1})}\prod_{j=1}^{m}\eta(\boldsymbol{P}_j). \tag{10.65}$$

Equation (10.64) suggests a Monte Carlo algorithm for evaluating G. Each particle history is created in the following manner. (1) The coordinates of the first collision, $\boldsymbol{P}_1$, are obtained by sampling from the first-flight collision density rate $\overline{S}(\boldsymbol{P}_1)$ (after renormalization of $\overline{S}(\boldsymbol{P}_0)$ to make it a proper PDF). (2) Then for $i = 1, 2, \ldots, m+1$, sample $\alpha(\boldsymbol{P}_i)$ to see if the history ends in an absorption at the ith collision. (3) Sample $\kappa(\boldsymbol{P}_{i-1}\to\boldsymbol{P}_i)$ to find the next collision coordinates (assuming the particles survived the previous collision and the chain is to be continued). With this procedure, the quantity

$$\big\{\overline{S}(\boldsymbol{P}_1)[1-\alpha(\boldsymbol{P}_1)]\kappa(\boldsymbol{P}_1\to\boldsymbol{P}_2)\ldots$$
$$\ldots[1-\alpha(\boldsymbol{P}_m)]\kappa(\boldsymbol{P}_m\to\boldsymbol{P}_{m+1})\alpha(\boldsymbol{P}_{m+1})\big\}\,d\boldsymbol{P}_1, d\boldsymbol{P}_2,\ldots,d\boldsymbol{P}_{m+1} \tag{10.66}$$

can be identified as the probability that the initial coordinates are in $d\boldsymbol{P}_1$ about $\boldsymbol{P}_1$, the second in $d\boldsymbol{P}_2$ about $\boldsymbol{P}_2$, and so on until the chain is terminated by an absorption at collision $m+1$.

The quantity $W(\boldsymbol{P}_1,\boldsymbol{P}_2,\ldots,\boldsymbol{P}_{m+1})$ is the contribution of the history to the tally for G. Moreover, from the interpretation of the expression inside the brackets of Eq. (10.66) as the joint PDF for $(\boldsymbol{P}_1,\ldots,\boldsymbol{P}_{m+1})$, it is seen that the right side of Eq. (10.64) is just the expected value of W, i.e., $\langle W\rangle = G$. Thus, sampling from the PDFs $\overline{S}$, α, and κ and scoring the resulting value of W defined by Eq. (10.65) yield an unbiased estimator for G. The Monte Carlo estimate of G is then obtained by sampling N histories, calculating W for each history, and averaging the scores, i.e.,

$$G \simeq \frac{1}{N}\sum_{i=1}^{N} W(\boldsymbol{P}_1^i,\ldots,\boldsymbol{P}_{n_i}^i),$$

where n_i is the collision number when the ith particle was absorbed or escaped.

10.11.2 The Integral Equation Method as Simulation

It may not be readily apparent how the above sampling scheme, based on the integral transport equation, is related to the Monte Carlo simulation method, outlined in Section 10.1. The simulation approach had no need for a transport equation. However, the above scheme is really an equivalent approach for transport simulations. Here it is shown that the scheme based on the integral transport equation for estimating the functional G can be cast into a Monte Carlo simulation whose histories are constructed much more physically.

For simplicity, it is assumed here that the medium is nonmultiplying, i.e., the medium can only absorb or scatter particles. For such a medium $\eta(\boldsymbol{P}) = 1$. A history begins by sampling the source function $S(\mathbf{r}, E, \boldsymbol{\Omega})$ (properly normalized) to select starting values $(\mathbf{r}_0, E_0, \boldsymbol{\Omega}_0)$. Then, the flight distance R to the first collision point $\mathbf{r}_1$ is found by sampling R from

$$\mu(\mathbf{r}_0, E_0) \exp\left\{-\int_0^R \mu(\mathbf{r}_0 - R'\boldsymbol{\Omega}_0, E_0)\, dR'\right\}$$

(again properly normalized) to obtain the first collision point $\mathbf{r}_1 = \mathbf{r}_0 + R\boldsymbol{\Omega}_0$.

The next step is to determine if the first collision is an absorption so that the history should be terminated. This is done by sampling from the absorption probability $\alpha(\mathbf{r}_1, E_0) = \mu_a(\mathbf{r}_1, E_0)/\mu(\mathbf{r}_1, E_0)$. However, as discussed later in Section 10.12.4, it is a bad idea to terminate a history when an absorption event occurs. Rather, it is more efficient to continue the particle history after reducing the weight of the particle by multiplying its weight by the nonabsorption probability $[1 - \alpha(\mathbf{r}_0, E_0)] = \mu_s(\mathbf{r}_0, E_0)/\mu(\mathbf{r}_0, E_0)$. In this manner, every interaction is deemed to be a scatter.

After this (perhaps forced) scatter, a new particle energy and direction are determined from the appropriate scattering distributions. The new energy is obtained by sampling from the PDF

$$\frac{\mu_s(\mathbf{r}_1, E_0 \to E)}{\int dE\, \mu_s(\mathbf{r}_1, E_0 \to E)},$$

and the new direction is found by sampling from the PDF

$$\frac{\mu_s(\mathbf{r}_0, E_0 \to E_1, \boldsymbol{\Omega}_0 \bullet \boldsymbol{\Omega})}{\int_{4\pi} \mu_s(\mathbf{r}_0, E_0 \to E_1, \boldsymbol{\Omega}_0 \bullet \boldsymbol{\Omega})}.$$

The above sampling procedure is continued to successive collision events until the particle weight falls below some predetermined minimum value (at which point, Russian roulette—see Section 10.12.3—is used to, perhaps, end the history) or until the particle leaves the phase space of interest in the problem.

When the particle history is ended, its contribution is added to the tally for G, and a new particle history is begun.

10.12 Variance Reduction and Nonanalog Methods

To define the efficiency of a Monte Carlo calculation, one must take into account both speed and variance. The speed is, to a large extent, determined by the hardware. Of course, the skill of the computer programmer and the operating system of the computer also determine speed. The emphasis in this section, however, is the variance of the result of a calculation and ways to minimize it.

As discussed in Chapter 5, the power of Monte Carlo depends on using many nonanalog techniques to reduce the variance of the estimator. In this section, some of the more useful variance reduction techniques for particle transport are briefly reviewed.

10.12.1 Importance Sampling

The use of importance sampling can greatly increase the flexibility of scoring and sampling procedures. For any single step in the Monte Carlo process involving the probability of survival of a particle or of its reaching a certain point, the result of the sampling process, insofar as its effect on the final answer is concerned, may be taken as the product of the particle weight and the probability of occurrence of the event. Thus, if a PDF $f(x)$ describes some random process, one is allowed to use some alternative, perhaps simpler, PDF $\tilde{f}(x)$; however, the ultimate expectation of the process must be left unchanged, and therefore the particle's weight before entering the process must be adjusted by the factor $w(x)\tilde{f}(x)$, where the weight function $w(x) \equiv f(x)/\tilde{f}(x)$. The function $\tilde{f}(x)$ is somewhat arbitrarily chosen, although it should not deviate too markedly from $f(x)$ and should have the nature of a proper PDF. In an analog simulation, the use of an alternative PDF instead of $f(x)$ is compensated by the subsequent testing for rejection or acceptance. In the importance sampling technique, the value of x selected is accepted without further ado, but the particle weight after the process is obtained by multiplying its previous weight by $w(x)$, where the argument is the value of x selected.

The use of importance sampling, which is a very effective variance reduction technique for any Monte Carlo problem, can also be used with great effect for particle transport. Consider the problem of Section 10.11.1 in which

$$G = \int d\boldsymbol{P}\, g(\boldsymbol{P})F(\boldsymbol{P}) \tag{10.67}$$

is to be estimated. The collision density $F(\boldsymbol{P})$ is given by the integral transport equation of Eq. (9.76)

$$F(\boldsymbol{P}) = \int d\boldsymbol{P}'\overline{K}(\boldsymbol{P}' \to \boldsymbol{P})F(\boldsymbol{P}') + \overline{S}(\boldsymbol{P}). \tag{10.68}$$

The function $g(\boldsymbol{P})$ can be viewed as the contribution to G from a collision at $\boldsymbol{P}$. In many problems, $g(\boldsymbol{P})$ is very small over large portions of phase space, and, to reduce the number of collisions in such regions, it is desirable to bias the samplings that generate the simulated particle histories so that more collisions occur in regions where $g(\boldsymbol{P})$ is large. Ideally, if one could find an importance function $I(\boldsymbol{P})$ that is proportional to the expected contribution to G from a particle at $\boldsymbol{P}$, this function could be used to bias the density functions used to generate the sequence of points in phase space that define a particle history.

The importance function can be used to define the following two functions:

$$\widetilde{S}(\boldsymbol{P}) \equiv \frac{\overline{S}(\boldsymbol{P})I(\boldsymbol{P})}{\int \overline{S}(\boldsymbol{P}')I(\boldsymbol{P}')\,d\boldsymbol{P}'}, \tag{10.69}$$

$$\widetilde{K}(\boldsymbol{P}' \to \boldsymbol{P}) \equiv \frac{\overline{K}(\boldsymbol{P}' \to \boldsymbol{P})I(\boldsymbol{P})}{I(\boldsymbol{P}')}. \tag{10.70}$$

As in Section 10.11.1, these functions can be used to construct biased histories with the weight of each particle adjusted at each step to compensate for the bias. Thus a source particle has its weight modified by a multiplicative factor of $\overline{S}(\boldsymbol{P})/\widetilde{S}(\boldsymbol{P})$, and each particle that enters a collision from a previous collision has its weight multiplied by $\overline{K}(\boldsymbol{P}' \to \boldsymbol{P})/\widetilde{K}(\boldsymbol{P}' \to \boldsymbol{P})$.

Now what function to pick for $I(\boldsymbol{P})$? It can be shown (Carter and Cashwell, 1975) that the optimum importance function I_{opt} is the solution of the equation

$$I_{opt}(\boldsymbol{P}) = \int d\boldsymbol{P}'\overline{K}(\boldsymbol{P} \to \boldsymbol{P}')I_{opt}(\boldsymbol{P}') + g(\boldsymbol{P}). \tag{10.71}$$

This is just the integral adjoint transport equation of Eq. (9.88) with the adjoint source specified as $\overline{S}^{\dagger}(\boldsymbol{P}) = g(\boldsymbol{P})$. With the optimum importance function, the Monte Carlo calculation gives a score of G for *every* history and, thus, is an estimator with zero variance! However, solving Eq. (10.71) is as much work as solving the original transport equation Eq. (10.68). Indeed, if $I_{opt}(\boldsymbol{P})$ were known, G could be estimated with simple numerical quadrature by using Eq. (9.92), namely,

$$G = \int g(\boldsymbol{P})F(\boldsymbol{P})\,d\boldsymbol{P} = \int \overline{S}(\boldsymbol{P})I_{opt}(\boldsymbol{P})\,d\boldsymbol{P}. \tag{10.72}$$

Although this optimum importance function is not known beforehand, its existence tempts one to try to develop various approximations for it without unduly increasing the computation time. These approximations are often based on deterministic methods such as discrete ordinate calculations using a simplified geometry (Tang et al., 1976). Alternatively, a Monte Carlo calculation of the adjoint flux can be performed using simplified physics models such as energy multigroup cross sections (MCNP, 2003).

10.12.2 Truncation Methods

During the course of its random walk, a particle may reach a location, direction, energy, or weight for which the likelihood of scoring is negligible. The energy may be so low or the position so remote that terminating the tracking of the particle is justified. Truncation is a particularly valuable tool in the tracking of charged particles, which have reasonably well-defined ranges, so that a truncation decision is straightforward. A special type of truncation usually associated with small particle weights is called *Russian roulette*. The Russian roulette method of truncation is addressed along with *splitting*, its companion procedure used for variance reduction.

10.12.3 Splitting and Russian Roulette

Suppose that by virtue of its position, direction, or some other feature, a particle is relatively likely to make an appreciable contribution to a score. The variance may be reduced by *splitting* the particle into two or more, say m, particles each having the same characteristics as the original particle, except each has a weight given by the fraction $1/m$ of the weight of the original particle. Each particle is then tracked independently. On the other hand, a particle may be relatively unlikely to make an appreciable contribution to a score. Then, a random number ρ may be selected, and if less than some fraction, say $1/m$, the particle may be *killed* by the Russian roulette scheme. Otherwise, tracking the particle would continue, but the particle's weight would be multiplied by the factor m.

Splitting and Russian roulette are normally used together. Different spatial or energy regions, for example, may be assigned *importances* I. As a particle moves from region j to region k, if $n \equiv I_k/I_j$ exceeds unity, the particle is split into n particles[11] with weights adjusted by the factor $1/n$. If n is less than unity, then Russian roulette may be played, with the particle surviving with probability $1 - n$ and its weight multiplied by the factor $1/n$. In a shielding problem with particles penetrating a thick-slab shield, Carter and Cashwell (1975) suggest that, for geometric splitting and Russian roulette, the slab be divided into layers of one mean free path thickness. As particles pass from region to region away from the entry face, they should be split 2 for 1. Russian roulette with 1/2 survival probability should be applied as particles pass from region to region toward the entrance face.

10.12.4 Implicit Absorption

An example of the usefulness of weighting is in the avoidance of the absorption process for a particle. Since particles absorbed in a medium give zero contribution to the final answer in many transport calculations, one intuitively has

[11] If n is not integer, the particle may be split into one of the bounding-integer number of particles and the weights adjusted as discussed in Section 5.8.

the feeling that a history terminating in this way is in a sense a waste of computer time. Thus, if a collision is deemed an absorption, the particle weight is reduced by multiplying its incident weight by the probability of survival, i.e., $[1-(\mu_a/\mu)]$, and the type of collision is redetermined by sampling from all the nonabsorption interaction probabilities. This numerical avoidance of absorption is sometimes called *implicit absorption* or *survival biasing*, as distinct from *analog absorption* in which the particle history is actually terminated.

However, in such a scheme the only way for a history to terminate is for the particle to leak from the system, and this can result in the buildup of a large number of particles with very low weights and that contribute little to the tally. This implicit capture is almost always combined with Russian roulette (see Section 10.12.3) to artificially reduce the population of low-weight particles when a particle's weight falls below some specified value.

10.12.5 Interaction Forcing

Consider, for example, the passage of photons or neutrons through important regions, but regions with dimensions of only one or so mean free paths, so that a particle is likely to pass through the region without interaction. The effort wasted by tracking a particle through such a region can be eliminated by *interaction forcing*. Suppose that the distance through the region along the particle's path is S; collisions may be forced if path lengths are selected by sampling the following alternative PDF:

$$\tilde{f}(s) = \frac{\mu e^{-\mu s}}{1-e^{-\mu S}}, \quad 0 \le s \le S. \tag{10.73}$$

The particle weight must then be adjusted by the factor $(1-e^{-\mu S})$.

10.12.6 Exponential Transformation

For Monte Carlo calculations involving penetration or reflection from a slab shield, there is motivation to artificially increase or decrease, respectively, the distance between collisions. To do this, the normal PDF for a sampling path length, $f(s) = \mu \exp(-\mu s)$, is modified by replacing μ by $(1-\beta\omega)\mu$, in which ω is the cosine of the angle between the preferred direction (toward the scoring region) and the direction of flight of the particle. The factor β is a biasing parameter, normally restricted to values less than unity. For deep penetration problems, this *exponential transformation* is often applied only to particles traveling in the generally favored direction (i.e., with $\omega > 0$). In other words, path lengths may be selected by sampling the following alternative PDF:

$$\tilde{f}(s) = (1-p\omega)\mu e^{-(1-\beta\omega)\mu s}. \tag{10.74}$$

For $\omega > 0$, the effective interaction coefficient $(1-p\omega)\mu$ is less than μ; thus, the particle path is longer than would be the case in a strictly analog calculation.

The corresponding weight factor is

$$w(s) = \frac{e^{-\beta\omega\mu s}}{1 - p\omega}. \tag{10.75}$$

By contrast, for $\omega < 0$, as may be the case for particles moving away from the scoring region, the effective interaction coefficient is greater than μ and the particle path is foreshortened. The parameter p must not be so near unity that excessively large weight factors result. In a particular application, it may be necessary to adjust p by trial and error to optimize calculations.

10.13 Summary

In a Monte Carlo neutral-particle transport simulation, the geometry of the system is first specified, typically by combinatorial geometry, although other approaches can be used. Then by sampling from many distributions, a complete simulation of a particle's track as it migrates through phase space can be obtained. Source sampling is required to pick the initial starting location, direction, and energy of a particle. Then sampling is used to pick a flight distance before a collision, followed by sampling to determine the type of collision. If the particle is not absorbed, more sampling is done to determine the type, direction, and energy of secondary particles. Each subsequent leg of the simulated track continues as for the first leg. The particle is tracked until it is absorbed or leaves the problem boundary. As a particle moves along its trajectory, various tallies are updated so that, after many histories, some desired properties of the radiation field can be estimated.

To reduce computational effort, transport simulations often change the physical sampling distributions to bias a particle's track to increase the chance of it scoring. To remove the bias from the score, a weight is assigned to each particle, and as the particle track evolves, the weight is adjusted to remove the bias each time a biased sampling is used. Such biasing can greatly reduce the variance of the tally and, consequently, most Monte Carlo simulations employ one or more variance reduction techniques.

With the advent of inexpensive and powerful personal computers or computer clusters and with the availability of powerful general-purpose Monte Carlo codes, Monte Carlo has become the method of choice for the vast majority of neutron and photon transport analyses.

Problems

10.1 Verify the functions $f_1(r)$ and $f_2(r)$ defined by Eq. (10.36) are ≤ 1 for $1 \leq r \leq 1 + 2/\lambda$.

10.2 Evaluate the constant K required to make Eq. (10.32) a properly normalized PDF. Then derive an expression for the efficiency of the sampling method described in Section 4.1.8. Plot the efficiency as a function of the incident photon energy.

10.3 In Monte Carlo particle transport, the distance s_{int} a particle must travel from a point $\mathbf{r}_o$ in direction $\boldsymbol{\Omega}$ to reach the nearest surface (in the direction of $\boldsymbol{\Omega}$) is of paramount importance if a particle is to be tracked as it moves through the problem geometry. Once s_{int} is known, the intersection point on the surface is readily found as $\mathbf{r}_{int} = \mathbf{r}_o + s_{int}\boldsymbol{\Omega}$. Derive formulas for s_{int} if $\mathbf{r}_o = (x_o, y_o, z_o)$ is inside (a) an infinite slab of thickness $2T$ centered at the origin, perpendicular to the y-axis, and parallel to the z-axis, (b) an infinite cylinder of radius R whose axis is the z-axis, and (c) a sphere of radius R centered on the origin.

10.4 Perform a Monte Carlo analysis, similar to that of Example 10.1, to determine the escape probability of a particle born uniformly in a purely absorbing sphere of radius R. Compare your results to the analytical result (Case et al., 1953)

$$P_{esc} = \frac{3}{8(\mu R)^3}\left[2(\mu R)^2 - 1 + (1 + 2\mu R)e^{-2\mu R}\right].$$

10.5 Perform a Monte Carlo analysis, similar to that of Example 10.1, to determine the escape probability of a particle born uniformly in a purely absorbing infinite cylinder of radius R. Compare your results to the analytical result (Case et al., 1953)

$$\begin{aligned} P_{esc} = \frac{2\mu R}{3}&\{2[\mu R\{K_1(\mu R)I_1(\mu R) + K_0(\mu R)I_0(\mu R)\} - 1] \\ &+K_1(\mu R)I_1(\mu R)/(\mu R) - K_0(\mu R)I_1(\mu R) + K_1(\mu R)I_0(\mu R)\}, \end{aligned}$$

where K_n and I_n are the nth-order modified Bessel functions of the first and second kind, respectively.

10.6 Verify Eqs. (10.19) to (10.21) by evaluating Eq. (10.17).

10.7 Derive the rotation matrix $\mathbf{R}_{xy}(\theta, \psi)$ for first a rotation about the x-axis followed by a rotation about the z-axis. Verify that the inverse of this matrix transforms the vector $\boldsymbol{\Omega}(\theta, \psi)$ in the vector $(0, 1, 0)$.

10.8 Consider a purely absorbing slab of thickness z mean free path lengths that is uniformly and normally illuminated by particles. The average flux density in the slab can be estimated analytically by (a) the path length estimator and (b) the reaction density estimator. Show that the relative error (ratio of the standard deviation to the mean) of these two estimators is as follows:

$$\text{collision estimator} = \sqrt{\frac{e^{-z}}{(1 - e^{-z})}},$$

$$\text{path length estimator} = \frac{\sqrt{1 - 2ze^{-z} - e^{-2z}}}{(1 - e^{-z})}.$$

(a) Plot these results as a function of z. (b) Also determine the limiting values as $z \to 0$ and as $z \to \infty$. (c) Discuss which estimator is best for optically thick and thick media and explain why these results are expected.

10.9 Write a code to produce the results of Example 10.2. Add to Fig. 10.15 results for $c = 0.5$ and $c = 0.9$.

10.10 Write a code to produce escape probabilities similar to those of Example 10.2 but for a sphere of radius R. (a) Verify your code by comparing the output for a very small c to the analytic expression given in Problem 10.4 for $c = 0$. (b) Besides using an analog tally $P_{esc} = N_{esc}/N_{total}$, i.e., the number of escaping particles to the total number of histories, add to your code a leakage estimator tally. For both tallies, the relative error of the estimated escape probabilities should be given.

10.11 This problem requires development of a computer code for calculation of transmission and reflection of photons normally incident (parallel to the x-axis) on a homogeneous shielding slab surrounded by a vacuum. It is a generalization of the problem of Example 10.3. For simplicity, assume one-speed particles and isotropic scattering.

(a) Calculate the reflected and transmitted angular flux densities. In particular, for $c = 0.9$, plot the angular distribution of the escaping angular flux density $\phi(0, \omega)$, $-1 < \omega < 0$, for $c = 0.9$ for different slab thicknesses.

(b) Estimate the reflected and transmitted currents $j_x(0, \omega)$ and $j_x(T, \omega)$. Plot the total reflected and transmitted leakage currents $j_x^-(0)$ and $j_x^+(T)$ (see Eq. (9.20)) as a function of slab thickness for various values of $c = \mu_s/\mu$.

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